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Greater China Semiconductors | Asia Pacific

Global AI ASICs Outlook,
Quantum Security, and China AI
compute

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS	
Asia Pacific	
Industry View	Attractive

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Strong AI Semi Outlook

- Top ideas:
 - **OW:**
 - **AI:** MediaTek (Top Pick), TSMC, SMIC, Aspeed, Alchip, KYEC, ASE, FOCI, ASMPT, AllRing, GUC
 - **Memory:** Macronix, AP Memory (Top Pick), Nanya Tech, Winbond, GigaDevice
 - **China – AI/Semis/WFE:** Iluvatar, Cambricon, Hygon, NAURA Tech, AMEC
 - **Testing Equipment and Consumables:** WinWay, MPI, Hon Hai Precision, Gudeng
 - **Mature nodes:** UMC
 - **EW:** OmniVision, Phison, MetaX, Realtek, GlobalWafers
 - **UW:** WIN Semi, Silergy, ASMedia

Long-term demand drivers:

- **Chip inflation:** We expect “price elasticity” to affect demand for tech products. Rising wafer, OSAT, and memory costs create more margin headwinds for chip designers in 2026.
- **AI cannibalization:** Besides demand weakness (AI replacing some human jobs), we see the semi supply chain also prioritizing AI semis over non-AI semis, e.g., T-Glass and memory shortages.
- **China – AI: DeepSeek is triggering inferencing AI demand:** DeepSeek has demonstrated cheaper inferencing. Local foundry supply chain is also becoming more capable in terms of AI GPU production.

Valuation Comparison: Foundry, Back-end, Memory, IDM and Semi Cap

							Dividend	FCF	Market	3M Avg. Daily	P/E Ratio (x)			EPS Growth			ROAE			P/B Ratio (x)		
Ticker	Company	Price 8/24/2026	Curr ency	Price Target	Upside/ Downside	Rating	Yield (%)	Yield (%)	Cap (US\$M)	Trading (US\$M)	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e
Foundry																						
2330.TW	TSMC	2,375.0	TWD	2,988.0	26%	O	1.4%	1.8%	1,944,455	2,813.1	21.4	16.4	12.9	67%	31%	27%	44%	41%	38%	7.9	5.7	4.2
2303.TW	UMC	123.5	TWD	138.0	12%	O	5.1%	5.3%	49,231	988.0	27.5	21.8	17.9	35%	26%	22%	14%	17%	19%	3.8	3.5	3.3
0981.HK	SMIC	66.8	HKD	85.0	27%	O	0.0%	-5.0%	76,261	1,294.0	NM	NM	NM	126%	65%	18%	7%	11%	11%	NM	NM	NM
5347.TWO	Vanguard	151.0	TWD	180.0	19%	E	4.9%	3.1%	8,861	154.8	27.6	22.3	18.2	31%	24%	23%	12%	14%	17%	3.3	3.1	2.9
1347.HK	Hua Hong	108.8	HKD	128.0	18%	E	0.1%	-4.6%	24,175	849.3	NM	NM	NM	491%	76%	30%	4%	7%	8%	NM	NM	NM
6770.TW	PSMC	70.8	TWD	111.0	57%	O	0.0%	5.6%	9,443	529.2	NM	NM	NM	-151%	301%	28%	4%	11%	12%	1.5	1.2	1.0
3105.TWO	WIN Semi	355.0	TWD	300.0	-15%	U	0.5%	-3.8%	2,264	316.9	NM	37.6	27.6	83%	30%	36%	7%	9%	12%	3.6	3.5	3.4
8086.TWO	AWSC	112.0	TWD	130.0	16%	U	1.1%	-3.9%	695	18.5	17.7	15.7	12.7	89%	13%	24%	14%	14%	16%	2.4	2.1	1.8
Mean:							1.6%	-0.2%			22.7	25.2	19.5	96%	71%	26%	13%	15%	16%	3.7	3.2	2.8
Median:							0.8%	-1.0%			24.5	21.8	17.9	75%	30%	25%	10%	13%	14%	3.4	3.3	3.1
Memory																						
603986.SS	Giga Device	381.7	CNY	750.0	97%	O	0.3%	4.1%	35,411	4,395.8	11.0	8.5	7.3	1302%	30%	15%	78%	57%	45%	5.9	3.8	2.7
2408.TW	Nanya Tech	501.0	TWD	580.0	16%	O	0.0%	13.6%	48,758	1,408.3	8.1	4.5	6.7	NM	NM	NM	77%	68%	29%	4.1	2.0	1.5
2344.TW	Winbond	177.0	TWD	288.0	63%	O	0.0%	16.6%	25,151	1,053.2	7.1	3.6	3.0	NM	NM	NM	68%	67%	46%	3.6	1.8	1.1
8299.TWO	Phison	2,135.0	TWD	2,588.0	21%	E	1.5%	1.6%	14,694	376.5	5.7	16.2	37.9	839%	-65%	-57%	83%	22%	10%	3.4	3.7	3.8
SMO.O	Silicon Motion	242.7	USD	400.0	65%	O	2.2%	2.1%	8,321	295.7	21.1	11.7	8.8	263%	81%	33%	38%	47%	45%	6.7	4.7	3.5
2337.TW	Macronix	124.0	TWD	220.0	77%	O	4.6%	17.2%	7,265	406.1	NM	NM	NM	-1421%	159%	5%	69%	81%	46%	2.6	1.1	0.7
Mean:							1.4%	9.2%			10.6	8.9	12.7	246%	51%	-1%	69%	57%	37%	4.4	2.9	2.2
Median:							0.9%	8.8%			8.1	8.5	7.3	551%	55%	10%	73%	62%	45%	3.9	2.9	2.1
Backend																						
3711.TW	ASEH	592.0	TWD	628.0	6%	O	2.2%	6.0%	81,563	486.0	33.1	19.7	12.7	94%	68%	55%	20%	28%	36%	6.0	5.0	4.0
600584.SS	JCET	73.8	CNY	50.0	-32%	E	0.3%	5.7%	18,346	2,566.2	51.1	36.9	31.0	65%	38%	19%	8%	11%	11%	4.1	3.7	3.4
601231.SS	USI	25.9	CNY	45.4	75%	O	1.4%	-7.4%	8,201	199.1	21.8	14.7	12.6	42%	48%	17%	13%	17%	17%	2.6	2.3	2.1
2449.TW	KYEC	239.0	TWD	338.0	41%	O	1.9%	1.6%	9,228	292.1	26.7	17.5	12.9	0%	53%	35%	21%	28%	32%	5.2	4.3	3.5
3363.TWO	FOCI	567.0	TWD	708.0	25%	O	0.2%	4.8%	1,883	68.5	NM	NM	NM	NM	NM	NM	-2%	55%	79%	NM	NM	NM
Mean:							1.5%	1.5%			24.3	16.1	12.8	50%	52%	31%	15%	21%	24%	4.5	3.8	3.2
Median:							1.7%	3.7%			24.3	16.1	12.8	54%	51%	27%	16%	22%	24%	4.7	4.0	3.4
Power Semis/ IDM																						
688396.SS	CR Micro	61.2	CNY	51.6	-16%	U	0.2%	1.6%	11,159	373.3	51.4	34.5	24.8	138%	49%	39%	6%	9%	11%	3.0	2.8	2.5
600460.SS	Silan Micro	35.6	CNY	26.9	-24%	U	0.5%	2.1%	8,221	685.8	43.1	29.0	22.6	244%	49%	28%	10%	14%	15%	4.2	3.7	3.3
603290.SS	StarPower	93.0	CNY	120.0	29%	E	1.0%	0.0%	3,094	185.8	59.4	34.9	23.3	-7%	70%	50%	5%	8%	12%	3.1	2.8	2.6
300373.SZ	Yangjie	97.8	CNY	136.0	39%	O	1.4%	-2.8%	7,336	425.6	32.6	23.1	17.7	29%	41%	31%	15%	19%	21%	4.7	4.0	3.3
Mean:							0.8%	0.2%			46.6	30.3	22.1	101%	52%	37%	9%	12%	15%	3.8	3.3	2.9
Median:							0.7%	0.8%			47.3	31.7	23.0	84%	49%	35%	8%	11%	14%	3.6	3.3	2.9
Semicap / Equipment/ Materials/ Supply Chain																						
002371.SZ	NAURA	707.9	CNY	818.0	16%	O	0.6%	-0.9%	71,262	1,378.7	55.7	38.1	29.9	45%	46%	27%	19%	24%	25%	10.0	8.3	6.8
688012.SS	AMEC	357.7	CNY	488.0	36%	O	0.0%	17.6%	31,208	1,631.8	51.4	46.4	34.1	116%	11%	36%	18%	16%	19%	8.3	7.0	5.8
7769.TW	Hon Precision	6,255.0	TWD	10,008.0	60%	O	2.5%	NA	35,533	152.4	48.4	28.1	16.8	69%	72%	67%	37%	53%	65%	NM	13.1	9.4
6223.TWO	MPI	5,565.0	TWD	8,000.0	44%	O	0.0%	-5.4%	17,218	186.4	86.3	37.0	17.4	98%	133%	113%	35%	57%	76%	NM	NM	NM
6515.TW	Winway	6,145.0	TWD	12,000.0	95%	O	0.9%	1.3%	6,937	103.2	NM	28.6	14.0	99%	132%	104%	44%	69%	84%	NM	NM	NM
6488.TWO	GlobalWafers	1,010.0	TWD	900.0	-11%	E	2.7%	1.7%	15,248	267.7	42.7	28.4	19.8	55%	50%	44%	12%	16%	21%	4.9	4.4	3.9
0522.HK	ASM Pacific	157.8	HKD	248.0	57%	O	1.8%	6.5%	8,023	101.9	35.9	23.1	16.8	73%	55%	38%	10%	15%	19%	3.5	3.2	2.9
688234.SS	SICC	105.1	CNY	127.1	21%	O	0.6%	NA	7,073	318.9	NM	NM	NM	NM	NM	NM	1%	3%	4%	7.0	6.8	6.6
ACMR.O	ACM Research	78.7	USD	130.0	65%	O	1.1%	-0.7%	5,165	131.1	27.8	18.0	13.7	104%	55%	31%	10%	14%	16%	2.5	2.2	1.9
3680.TWO	Gudeng Precision	464.0	TWD	648.0	40%	O	0.0%	-6.1%	1,407	27.3	23.9	16.9	12.2	106%	42%	38%	15%	19%	22%	3.4	2.9	2.4
6187.TWO	AllRing Tech	1,205.0	TWD	1,580.0	31%	O	2.8%	1.1%	3,661	95.4	47.4	30.3	26.8	64%	56%	13%	31%	39%	35%	NM	NM	NM
Mean:							1.2%	1.7%			46.6	29.5	20.2	83%	65%	51%	21%	29%	35%	5.6	6.0	5.0
Median:							0.9%	1.1%			47.4	28.5	17.1	86%	55%	38%	18%	19%	22%	4.9	5.6	4.9

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Data as of August 24, 2026.

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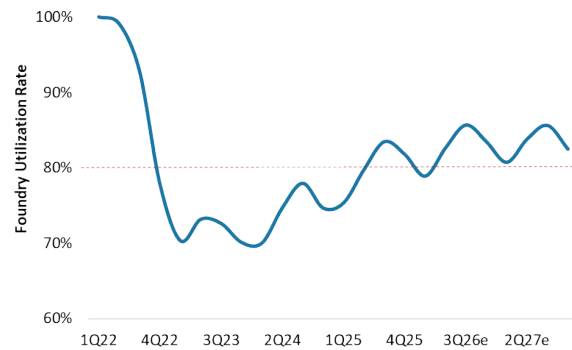
Valuation Comparison: Fabless, Power Semis, FPGA and Analog

Ticker	Company	Price 8/24/2026	Curr ency	Price Target	Upside/ Downside	Rating	Dividend Yield (%)	FCF Yield (%)	Market Cap (US\$M)	3M Avg. Daily Trading (US\$M)	P/E Ratio (x)			EPS Growth			ROAE			P/B Ratio (x)		
											2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e
Fabless																						
AI accelerators																						
688256.SS	Cambricon	969.1	RMB	1,408.0	45%	O	0.0%	-2.2%	84,580	2,625.0	84.9	51.0	39.4	134%	67%	29%	44%	52%	44%	33.9	21.4	14.7
9903.HK	Iluvatar	361.0	HKD	688.0	91%	O	NA	NA	11,740	153.3	NM	56.1	31.4	-83%	-805%	78%	-6%	26%	34%	16.8	12.9	9.2
688802.SS	MetaX	664.0	RMB	758.0	14%	E	0.0%	NA	36,908	244.4	NM	478.7	266.4	-103%	2214%	80%	0%	4%	7%	20.1	19.3	18.0
Smartphone/Wireless Semis																						
2454.TW	MediaTek	3,765.0	TWD	5,588.0	48%	O	17.3%	0.9%	186,079	1,475.6	51.5	25.6	12.2	17%	101%	110%	26%	39%	55%	12.2	8.4	5.5
603501.SS	OmniVision	82.5	CNY	111.0	35%	E	0.5%	2.1%	13,821	419.5	24.8	17.0	13.9	0%	45%	23%	14%	18%	19%	3.2	2.7	2.3
300782.SZ	Maxscend	74.6	CNY	80.00	7%	U	0.1%	-2.2%	5,529	285.2	NM	81.7	44.1	-20%	-309%	85%	-2%	5%	9%	4.1	3.9	3.6
603160.SS	Goodix	47.3	CNY	58.0	23%	U	0.7%	-1.4%	3,025	60.7	39.6	35.0	31.7	-35%	13%	11%	6%	6%	7%	2.3	2.2	2.0
6462.TWO	Egis	98.9	TWD	105.0	6%	E	1.0%	-16.3%	285	5.1	NM	18.4	NM	-88%	-462%	NA	-1%	5%	NA	1.0	1.0	NA
IP/Design Service																						
3661.TW	Alchip	3,735.0	TWD	5,888.0	58%	O	1.0%	5.1%	9,616	268.8	21.1	16.5	10.3	161%	28%	61%	32%	33%	44%	NM	NM	NM
301269.SZ	Empyrean Technology	91.3	CNY	95.0	4%	E	0.1%	NA	6,914	266.0	NM	NM	NM	283%	136%	70%	4%	10%	15%	NM	NM	NM
3443.TW	GUC	5,365.0	TWD	6,288.0	17%	O	0.8%	4.7%	22,703	346.1	107.4	38.8	25.6	79%	176%	52%	44%	74%	68%	NM	NM	NM
6531.TW	AP Memory	845.0	TWD	1,666.0	97%	O	1.2%	4.1%	4,318	128.7	47.8	23.9	13.0	129%	100%	84%	22%	36%	49%	9.6	7.6	5.5
PC/Cloud/Display Semis																						
3034.TW	Novatek	542.0	TWD	414.0	-24%	U	6.1%	14.3%	10,414	89.4	15.7	15.3	18.0	29%	2%	-15%	30%	28%	23%	4.4	4.2	4.2
2379.TW	Realtek	716.0	TWD	717.0	0%	E	4.8%	5.0%	11,535	119.9	19.1	19.1	16.4	30%	0%	16%	34%	31%	34%	6.2	5.8	5.3
5269.TW	ASMedia	1,400.0	TWD	1,482.0	6%	U	4.1%	0.9%	3,301	50.8	15.2	15.7	14.8	25%	-3%	6%	18%	16%	15%	2.6	2.3	2.1
5274.TWO	Aspeed	15,115.0	TWD	20,888.0	38%	O	1.3%	1.3%	18,043	134.1	78.7	55.7	38.3	85%	41%	45%	83%	81%	82%	NM	NM	NM
4966.TWO	Parade	557.0	TWD	818.0	47%	O	2.9%	1.6%	1,369	35.2	18.4	12.9	11.1	-11%	42%	17%	10%	14%	15%	1.9	1.7	1.6
2458.TW	Elan Micro	139.0	TWD	250.0	80%	O	4.8%	9.6%	1,257	25.9	13.3	11.3	11.2	24%	18%	1%	27%	28%	26%	3.4	3.0	2.8
HIMX.O	Himax	13.4	TWD	17.4	30%	E	2.7%	8.3%	2,283	41.9	33.6	18.5	9.2	65%	82%	102%	8%	14%	26%	2.7	2.5	2.2
Analog IC																						
6415.TW	Silergy	450.0	TWD	388.0	-14%	U	1.4%	4.8%	5,514	83.2	44.4	27.3	21.2	59%	62%	29%	10%	15%	17%	4.3	3.8	3.4
300661.SZ	SG Micro	111.1	CNY	82.0	-26%	E	0.3%	1.1%	9,574	341.2	81.5	57.1	45.8	49%	43%	25%	15%	18%	20%	NM	NM	NM
MCU																						
4919.TW	Nuvoton	116.0	TWD	95.0	-18%	U	0.8%	14.5%	1,538	58.9	NM	27.9	15.2	-70%	-455%	84%	-4%	13%	20%	3.9	3.4	2.8
688018.SS	Espressif	108.3	CNY	125.0	15%	O	NA	NA	3,504	137.4	NM	NM	NM	NA	NA	NA	12%	14%	15%	NM	NM	NM
FPGA																						
002049.SZ	Unigroup Guoxin	61.0	CNY	52.0	-15%	U	0.4%	1.5%	7,199	428.1	26.0	20.1	17.9	38%	29%	13%	14%	15%	15%	3.3	2.9	2.5
1385.HK	Shanghai Fudan	25.3	HKD	42.0	66%	O	0.1%	0.1%	6,905	21.4	17.5	17.8	16.1	415%	-2%	11%	18%	15%	14%	2.8	2.5	2.1
IC distributor																						
3702.TW	WPG	111.0	TWD	160.0	44%	O	7.2%	-27.2%	5,885	46.9	8.7	8.9	8.4	115%	-2%	5%	24%	21%	20%	1.8	1.7	1.6
3036.TW	WT Micro	200.5	TWD	349.0	74%	O	4.4%	-21.4%	8,012	82.8	8.1	7.1	6.3	107%	13%	13%	25%	25%	24%	1.9	1.7	1.5

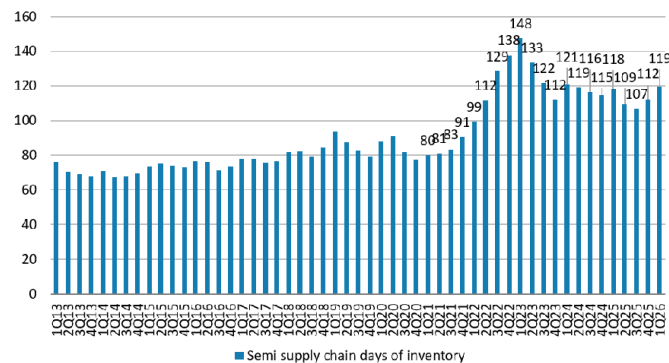
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates. Note: Data as of August 24, 2026.

Broader Semiconductor Cycle

Logic semi foundry utilization could reach 80% in 2H26

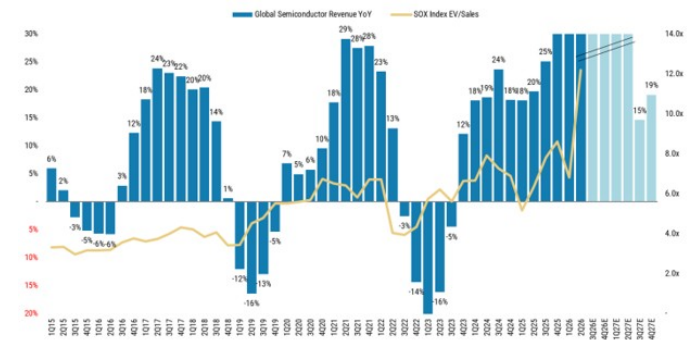


Semi supply chain – days of inventory as of 1Q26

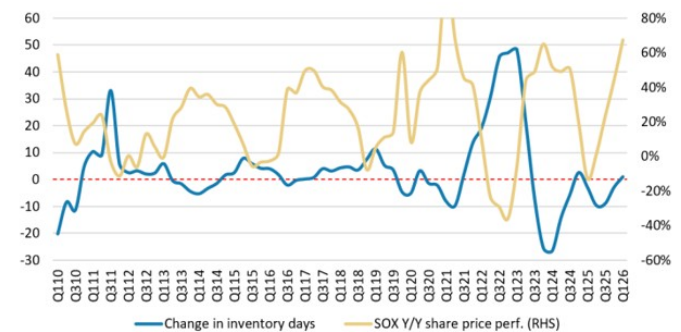


Source: SIA, FactSet, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Excluding memory and NVIDIA's AI GPU revenue, non-AI semi growth is expected to decline in 2026



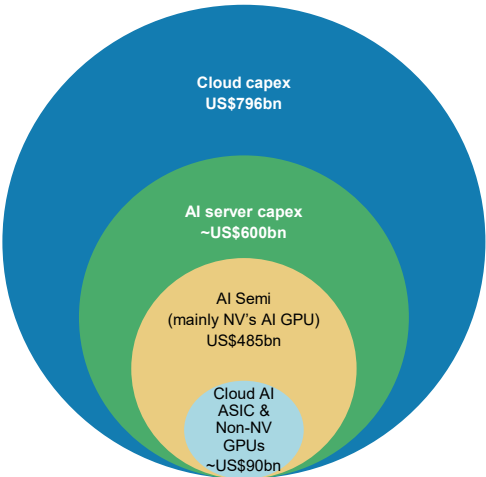
Historically, when inventory days have gone down, the semi stock index has risen



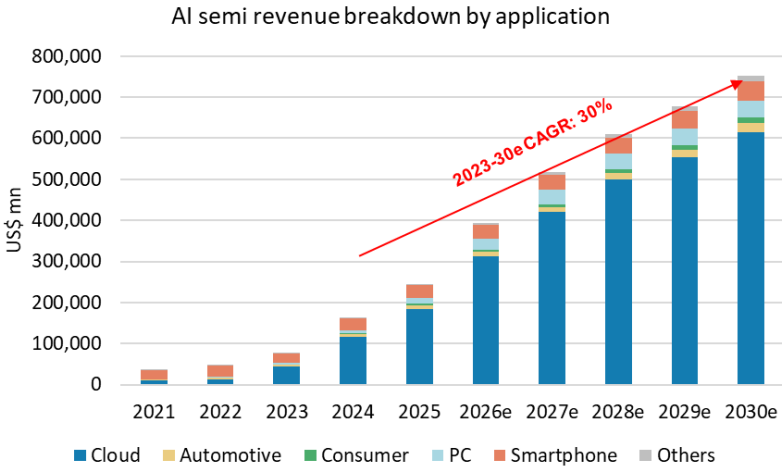
AI Semi Supply Chain and Niche Memory

Global Semi Industry Market Could Reach US\$1.5tn by 2030e, with AI Semis Contributing Half

Our supply chain data-driven bull case assumption is that cloud AI Semi TAM could grow to US\$485bn in 2026e



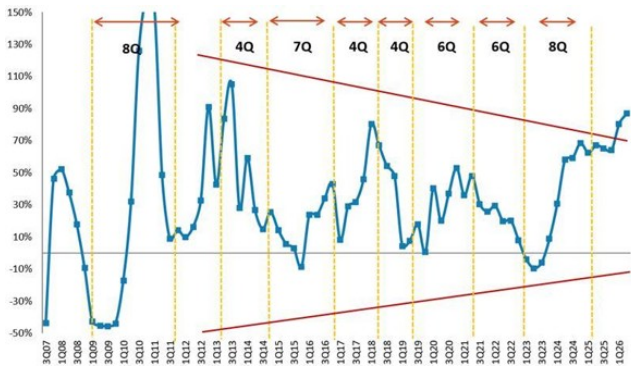
We expect AI semi TAM to reach ~US\$753bn by 2030e



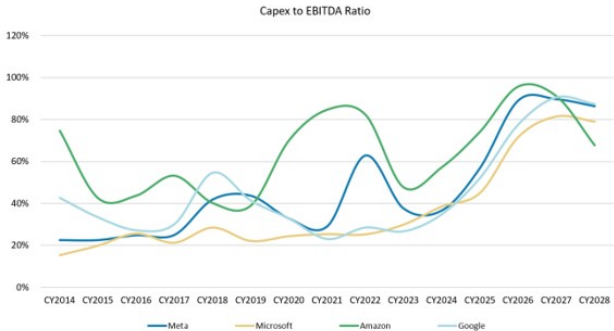
Source: Nvidia, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Cloud Semis: Brighter Outlook

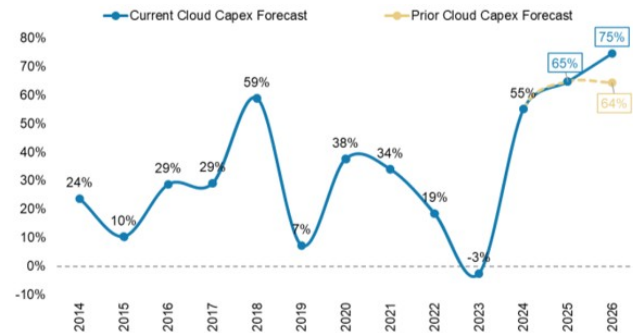
Top 4 CSPs (Amazon, Google, Microsoft and Meta): capex was up 87% Y/Y in 2Q26CY



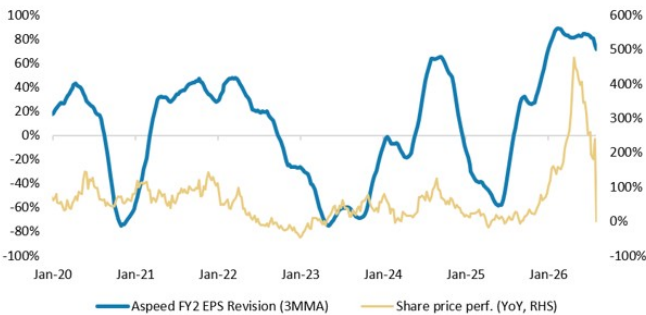
Capex-to-EBITDA ratio is reaching over 70%



Top 4 Cloud Providers: Cloud Capex Y/Y Growth



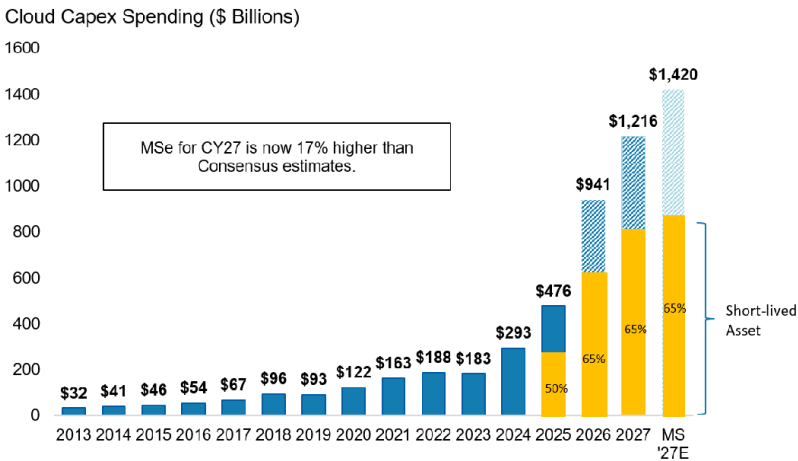
Aspeed – earnings estimate revision breadth



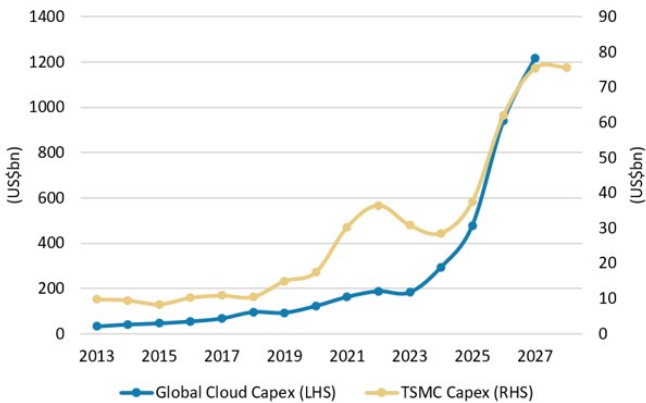
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Cloud Capex Remains Robust from Major CSPs

Morgan Stanley cloud capex tracker: we estimate nearly US\$1.4tn of cloud capex in 2027 (Top 14 listed global CSPs; no sovereign AI)

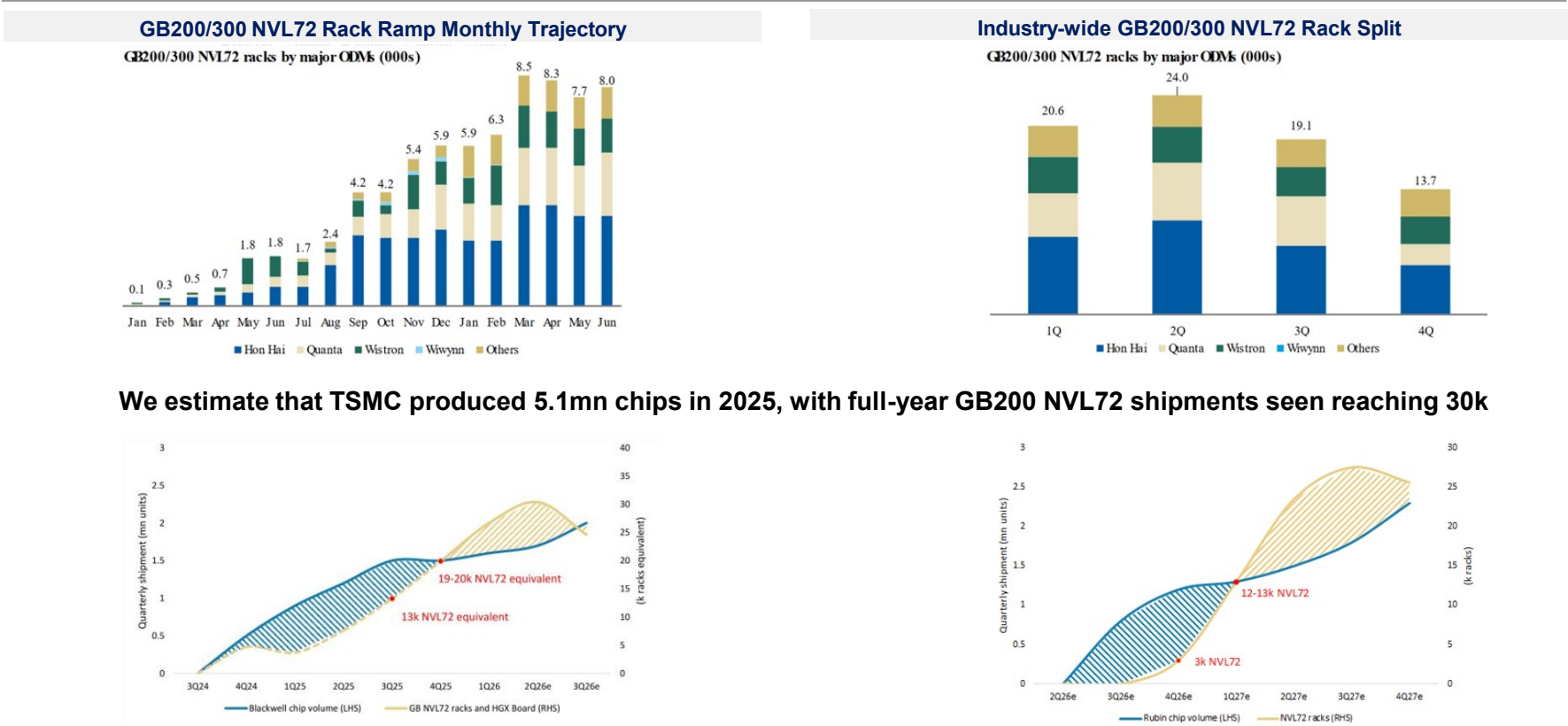


Global Cloud Capex vs. TSMC’s Capex



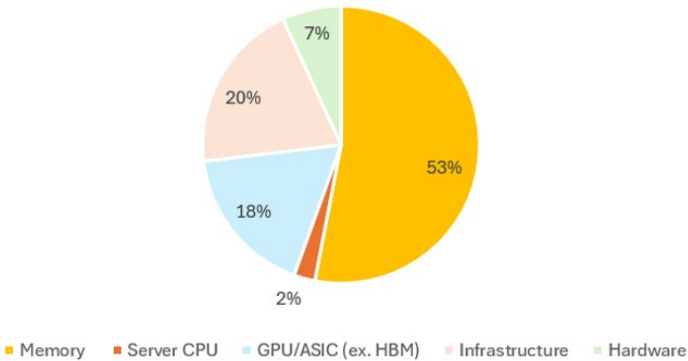
Source: FactSet, company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are top-down and from our US tech team.

NVIDIA NVL72 Rack Forecasts

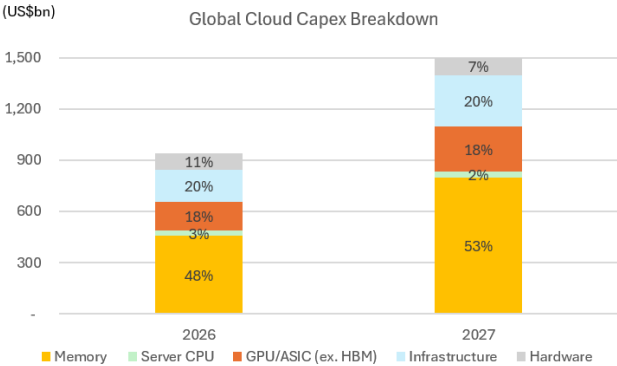


Logic Cloud Account for Around 20% of Global Cloud Capex

2027 Cloud Capex Breakdown



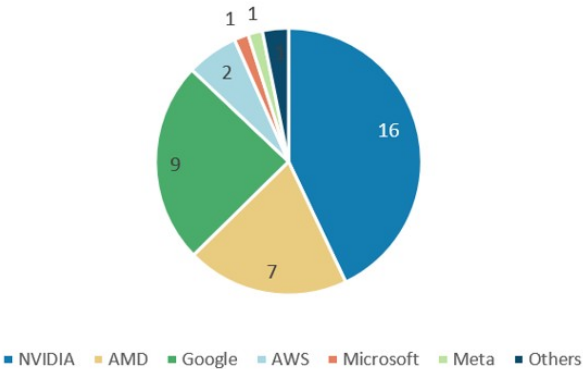
Global Cloud Capex Breakdown



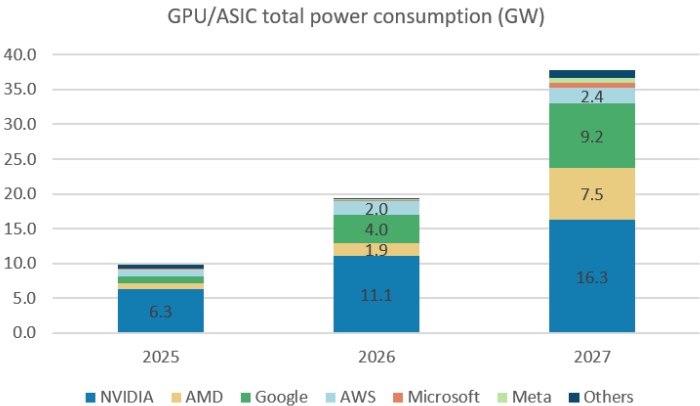
Source: Company data, Morgan Stanley Research estimates

Power Consumption Growth Remains Robust

2027 GPU/ASIC total power consumption (GW)



Global GPU/ASIC Total Power Consumption from TSMC CoWoS Shipment



Source: Company data, Morgan Stanley Research estimates

Morgan Stanley

Announced Power Deployment – Implications for TSMC

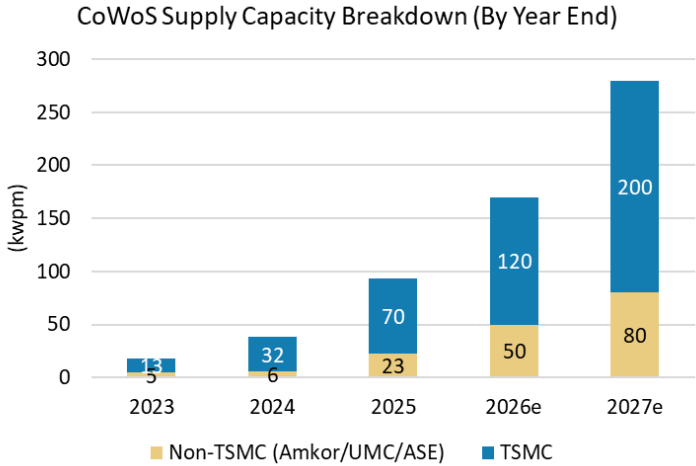
Announced Power Deployment Implication for TSMC					
GPU/ASIC Vendor	NVIDIA (Top US AI firm)	AMD (Top US AI firm)	Broadcom	AWS (Top US AI firm)	NVIDIA (SpaceX)
Power Deployed (GW)	10	6	3.5	2	8
Rack name	Vera Rubin NVL144	Helios	TPU	Trainium3 UltraServers	Vera Rubin NVL144
Power consumption per Rack (kW)	220	220	63	144	220
Rack Number (k units)	45	27	56	14	36
Chip Name	Rubin GPU	MI455 GPU	TPUv7 (Ironwood) TPUv8i (Sunfish)	Trainium3 (4)	Rubin GPU
Chip Volume (k units)	3,273	1,964	3,571	2,000	2,618
Life Cycle for the project	3	3	5	8	1
Implied life-cycle CoWoS Volume (k wafers)	409	166	260	118	409
Implied life-cycle 2/3nm wafer Volume (k wafers)	260	95	190	107	208
2027 annual CoWoS demand (k wafers)	136	55	52	15	409

Source: Company data, Morgan Stanley Research; Note: Estimates are compiled using our Asia supply-chain checks. Note: We assume 64 units of TPU chips per rack with TPUv7 and v8i per chip TDP being ~980W.

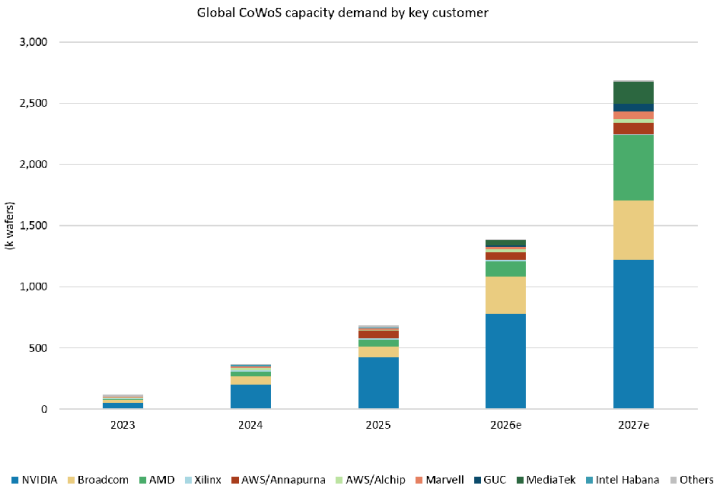
12

TSMC May Expand CoWoS Capacity to 200kwpm by 2027 Given Ongoing Strong AI Demand

Global CoWoS capacity expansion by year-end and by vendor



Global CoWoS consumption, by customer



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Global CoWoS Demand Breakdown

(k wafer)	2023	2024	2025	2026e	2027e	2023	2024	2025e	2026e	2027e
NVIDIA	53	200	425	780	1,222	45%	54%	62%	56%	45%
TSMC			390	680	1,090					
CoWoS-L			390	650	910					
CoWoS-S			0	20	50					
CoWoS-R			0	10	130					
Non-TSMC			35	100	132					
Amkor			35	100	132					
CoWoS-S			20	20	12					
CoWoS-R			15	80	120					
ASE/SPIL			0	0	0					
CoWoS-S			0	0	0					
CoWoS-R			0	0	0					
Broadcom	23	68	85	300	484	20%	18%	12%	22%	18%
TSMC			83	260	420					
CoWoS-L			0	15	55					
CoWoS-S			83	245	365					
ASE/SPIL			2	30	40					
CoWoS-S			2	30	40					
Amkor				10	24					
CoWoS-S				10	24					
AMD	7	40	60	130	530	6%	11%	9%	9%	20%
TSMC			60	80	240					
CoWoS-L			0	70	230					
CoWoS-S			60	10	10					
ASE/SPIL			0	50	170					
CoWoS-L			0	50	170					
Amkor				0	120					
CoWoS-L				0	120					

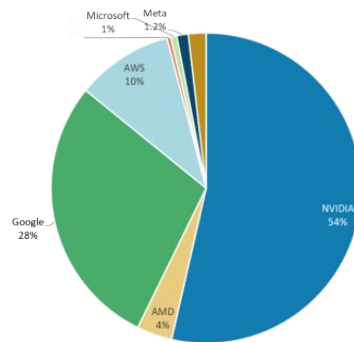
(k wafer)	2023	2024	2025	2026e	2027e	2023	2024	2025e	2026e	2027e
Xilinx	3	10	10	10	10	3%	3%	1%	1%	0%
MediaTek				40	180				3%	7%
TSMC				40	180					
CoWoS-S				40	180					
AWS/Annapurna			60	62	90				4%	3%
TSMC			60	32	54					
CoWoS-R			60	32	54					
ASE/SPIL				30	36					
CoWoS-R				30	36					
AWS/Alchip	9	16	5	26	36	8%	4%	1%	2%	1%
Intel Habana	0	7	9	0	0	0%	2%	1%	0%	0%
Marvell	1	18	15	17	64	1%	5%	2%	1%	2%
TSMC				5	50					
CoWoS-L				5	50					
CoWoS-R			15	0	0					
ASE/SPIL				12	14					
CoWoS-R				12	14					
GUC	1	1	2	14	60	1%	0%	0%	1%	2%
Cisco		2	3	5	6		1%	0%	0%	0%
Others	20	10	15	10	12	17%	3%	2%	1%	0%
Total demand	117	372	689	1,394	2,694	100%	100%	100%	100%	100%

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

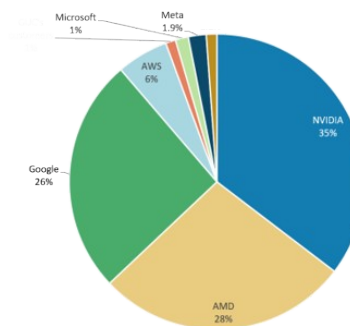
Morgan Stanley

AI Computing Wafer Consumption Could Reach >US\$46bn in 2027, with NVIDIA Accounting for the Majority

AI semi wafer consumption, by customer, 2026e



AI semi wafer consumption, by customer, 2027e

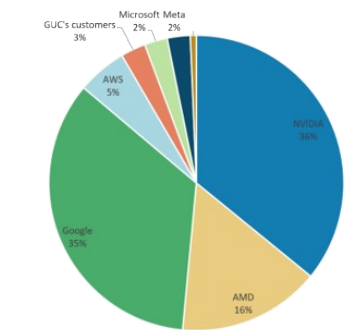


AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	Compute die size	Geometry	Compute die units	Wafer consumption (k wafers)	Wafer price (US\$)	Wafer revenue TAM (US\$ mn)
AI GPU (2027e)										
NVIDIA	B300	40	14	560	850	4nm	2	44	23,042	1,024
	Vera CPU	250	23	5,750	850	3nm	1	228	27,300	6,229
	Rubin R200	740	8	5,920	850	3nm	2	470	27,300	12,827
	Rubin Ultra	130	8	1,040	850	3nm	2	58	27,300	1,588
AMD	MI350 series	24	12	288	110	3nm	8	8	27,300	229
	MI455	157	7	1,099	850	2nm	2	13	30,000	400
	MI450 (for Meta)	35	14	490	850	2nm	1	3	30,000	89
	MI500 (Arcadia)	24	4	96		2nm			30,000	
	Venice CPU	270	21	5,670		2nm				
AI ASIC (2027e)										
Google	TPU v7p (Ironwood; AVGO)	13	16	208	700	3nm	2	14	27,300	371
	TPU v8i (Sunfish; AVGO)	330	12	3,960	800	3nm	2	296	27,300	8,075
	TPU v8t (Zebrafish; MediaTek)	180	20	3,600	800	3nm	2	269	27,300	7,341
	TPU v9 (Humafish; MediaTek)			400	850	2nm	2	32	27,300	867
AWS	Trainium 3	140	17	2,380	700	3nm	2	127	27,300	3,465
GUC's customers		60	20	1,200	645	4nm	1	29	23,042	674
Microsoft	Maia 300	50	11	550	850	2nm	1	29.1	30,000	873
Meta	MTIA 3 (Iris)	55	10	550	850	3nm	2	44	27,300	1,192
Others		18								
Total		2,664						1,700		46,208

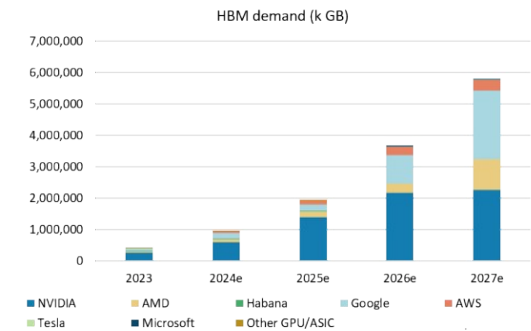
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are compiled using our supply-chain checks.

HBM Consumption in 2027e – Up to 48bn Gb

HBM consumption by customer in 2027e



NVIDIA still consumes most HBM supply in 2027e

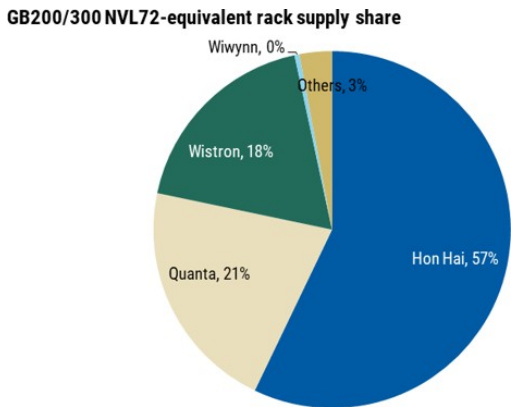


General assumptions				HBM assumptions						
AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	HBM chip density (GB)	HBM chip units	Total HBM size (GB)	HBM generation	HBM vendor	Total HBM demand (k GB)
AI GPU (2027e)										
NVIDIA	B300	40	14	560	36	8	288	HBM3e 12hi	Hynix/Micron	161,280
	Vera CPU	250	23	5,750	36	8	288	HBM4 12hi	Hynix/Micron/Samsung	1,704,960
	Rubin R200	740	8	5,920	36	8	288	HBM4e 12hi	Hynix/Micron/Samsung	399,360
	Rubin Ultra	130	8	1,040	48	8	384	HBM4e 12hi	Hynix/Micron/Samsung	399,360
AMD	MI350 series	24	12	288	36	8	288	HBM3e 12hi	Samsung/Micron	82,944
	MI455	157	7	1,099	36	12	432	HBM4 12hi	Samsung/Micron	474,768
	MI450 (for Meta)	35	14	490	36	6	216	HBM4 12hi	Samsung/Micron	105,840
	MI500 (Arcadia)	24	4	96	48	16	768	HBM4e 12hi	Samsung/Micron	73,728
	Venice CPU	270	21	5,670						
AI ASIC (2027e)										
Google	TPU v7p (Ironwood; AVGO)	13	16	208	24	8	192	HBM3e 8hi	Hynix/Samsung	39,936
	TPU v8i (Sunfish; AVGO)	330	12	3,960	36	8	288	HBM3e 12hi	Samsung/Hynix/Micron	1,140,480
	TPU v8t (Zebrafish; MediaTek)	180	20	3,600	36	6	216	HBM3e 12hi	Samsung/Hynix/Micron	777,600
	TPU v9 (Humafish; MediaTek)			400	48	12	576	HBM4e 12hi	Samsung/Hynix/Micron	230,400
AWS	Trainium 3	140	17	2,380	36	4	144	HBM3e 12hi	Hynix/Samsung/Micron	342,720
GUC's customers		60	20	1,200	24	6	144	HBM3e 8hi	Samsung?	172,800
Microsoft	Maia 300	50	11	550	36	8	288	HBM4 12hi	Samsung	158,400
Meta	MTIA 3 (Iris)	55	10	550	36	8	288	HBM3e 12hi	Samsung/Hynix/Micron	158,400
Others		18								
Total		2,664								6,077,256
Total HBM demand (mn Gb)										48,618

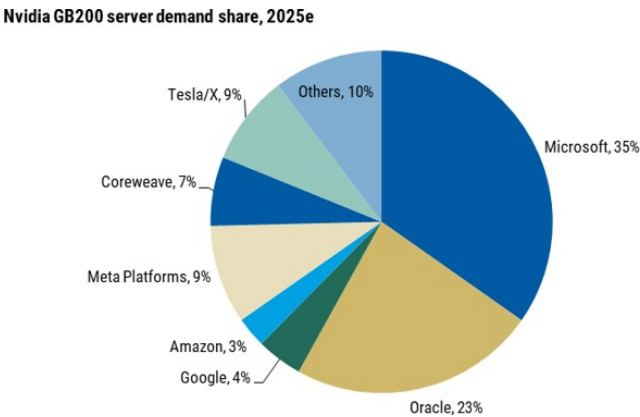
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

NVIDIA GB200/300 Server Racks – Supply/Demand Assumptions

NVIDIA GB200/300 rack supply share (2025)

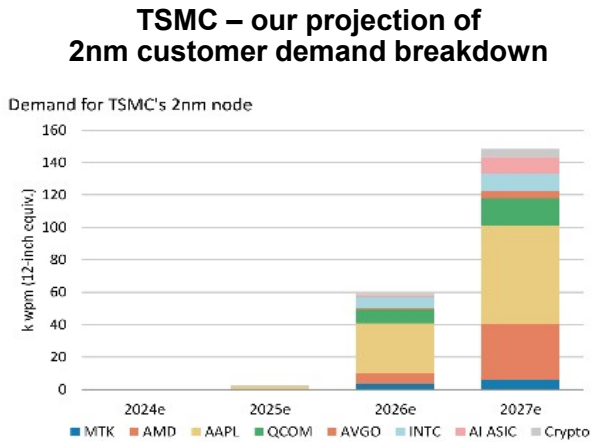


NVIDIA GB200/300 rack demand share (2025)

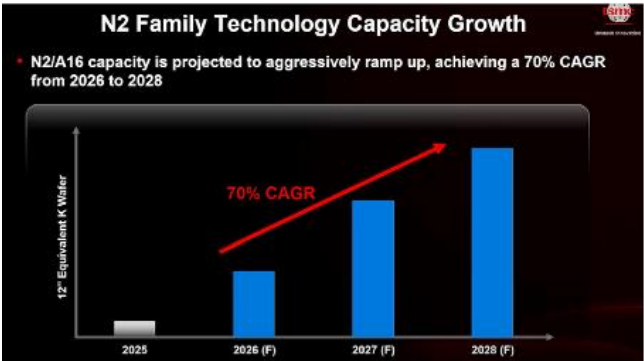


Source: Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are from our Asia Pacific tech team’s downstream estimates.

TSMC 2nm Demand Remains Very Strong

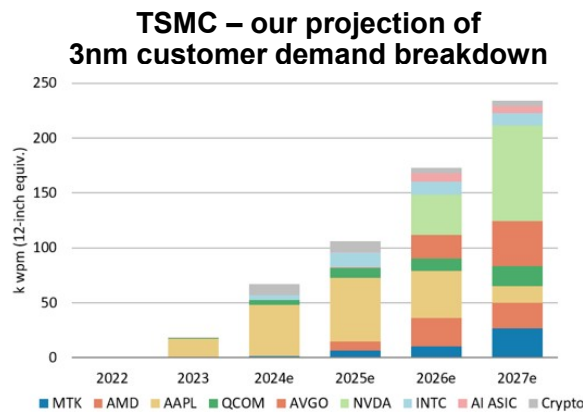


TSMC suggests N2 capacity to achieve 70% CAGR from 2026-2028e

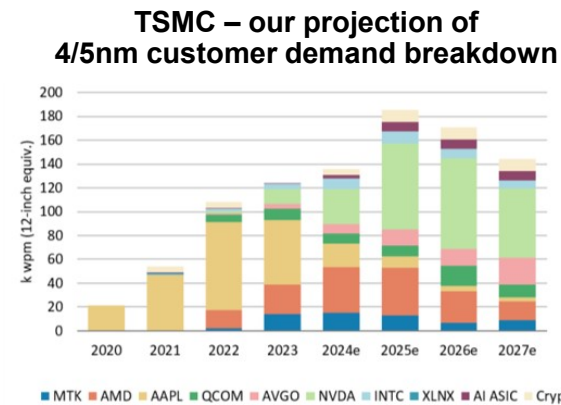
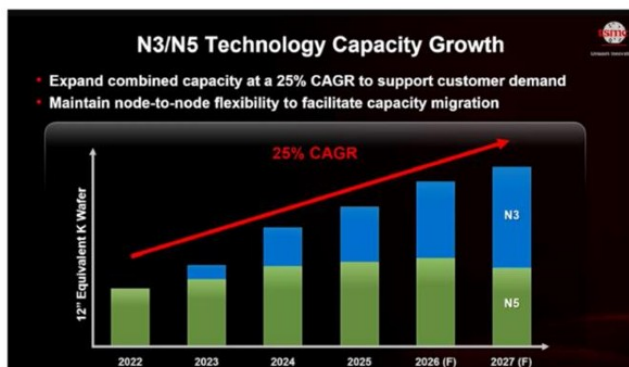


Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

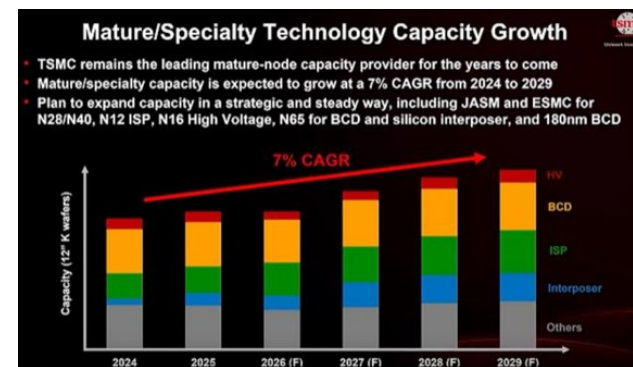
TSMC N5 Capacity to Drop in 2027 while N3 Should Keep Increasing, Similar to our Demand Forecast



TSMC indicates that N5 capacity will drop in 2027 while N3 capacity should keep increasing



TSMC mature/specialty capacity is expected to grow 7% CAGR from 2024-2029

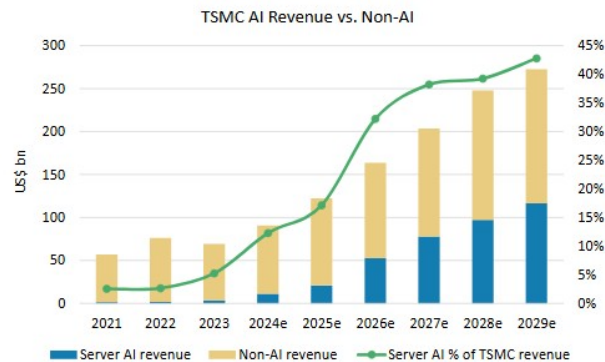


Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

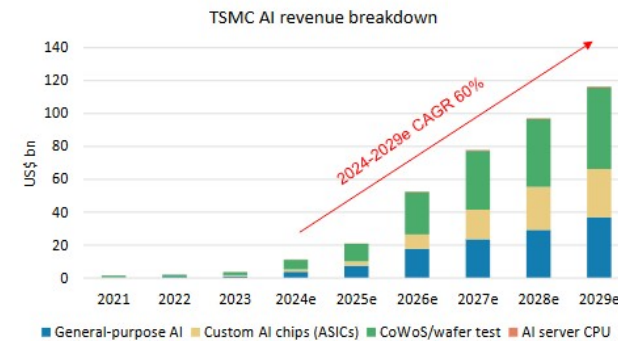
Morgan Stanley

TSMC – AI Semis Revenue Could Reach 60% from 2024 to 2029e

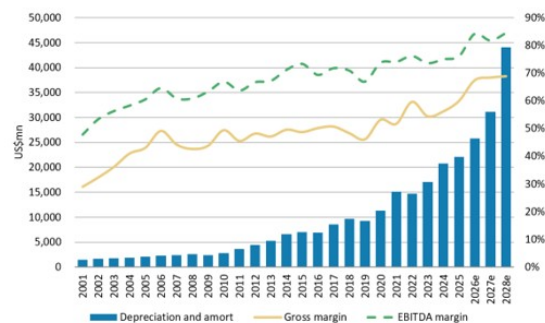
TSMC – AI semis revenue could account for >30% of 2026e revenue



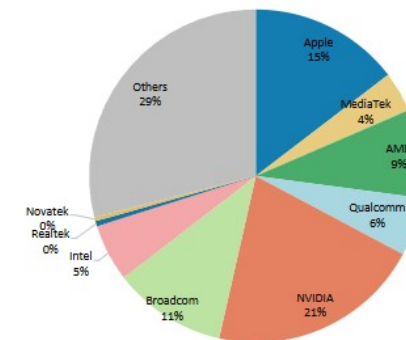
TSMC – AI semis revenue breakdown



TSMC – margin expansion



TSMC – 2026e revenue breakdown by customer



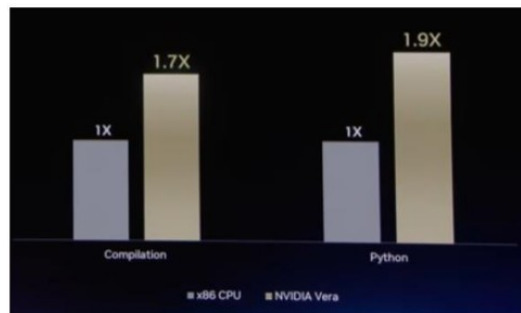
Source: Company data, Morgan Stanley Research (e) estimates

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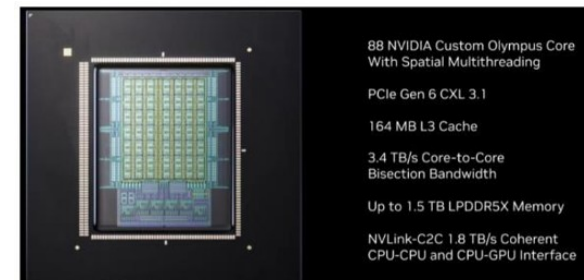
Morgan Stanley

Nvidia's Latest Vera CPU Could Support Robust Agentic CPU Demand

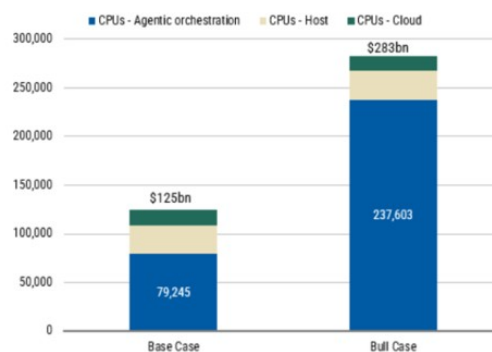
Vera performance could be 1.8x that of highest-performance x86 CPU



Vera CPU cores are not split across chiplets, enabling faster core-to-core connection



Our bull case implies a US\$238bn CPU orchestration TAM



In MS top-down model, we estimate the Agentic CPU TAM will expand at a 251% CAGR from F26 to F30

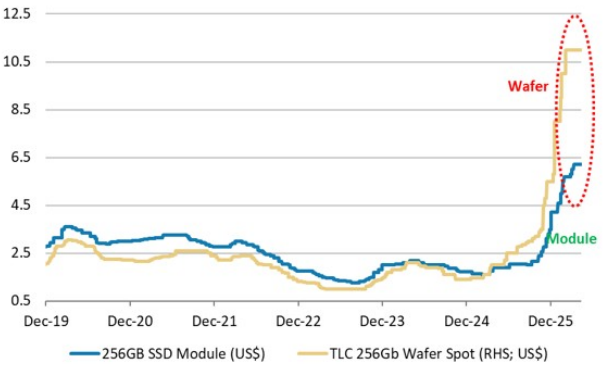


Source: Nvidia, Morgan Stanley Research (e) estimates from our Global team; see [Global Technology: Agentic AI – The Surge Begins \(11 May 2026\)](#)

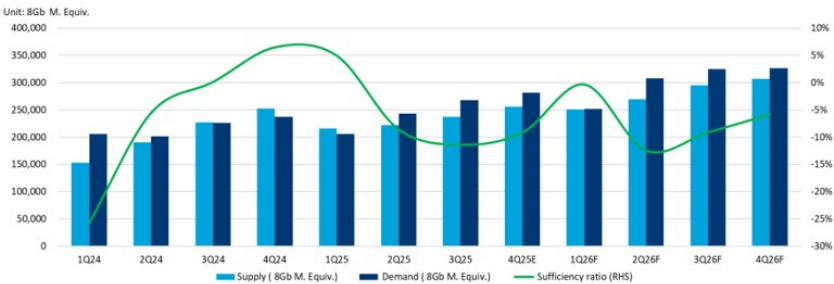
21

AI Storage Leads to NAND Shortage; We Also Expect NOR Flash Undersupply into 2026

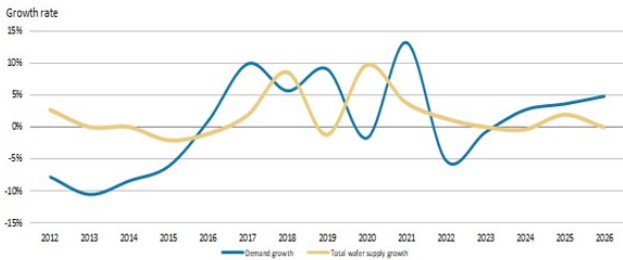
NAND wafer spot pricing and module prices



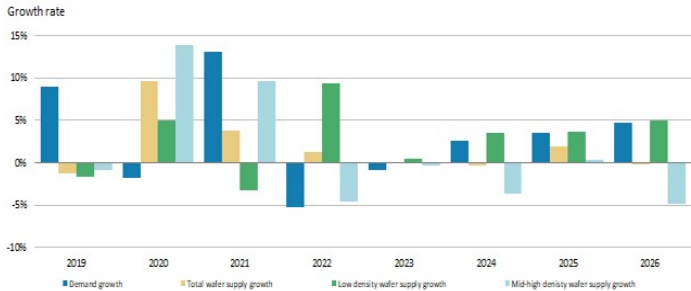
NAND supply and demand model



NOR flash demand and supply growth rates



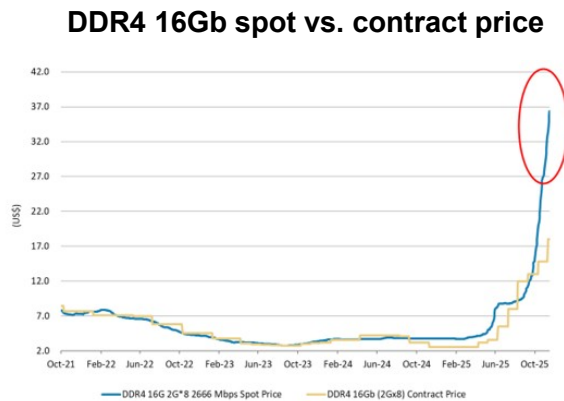
NOR flash demand and supply growth rates



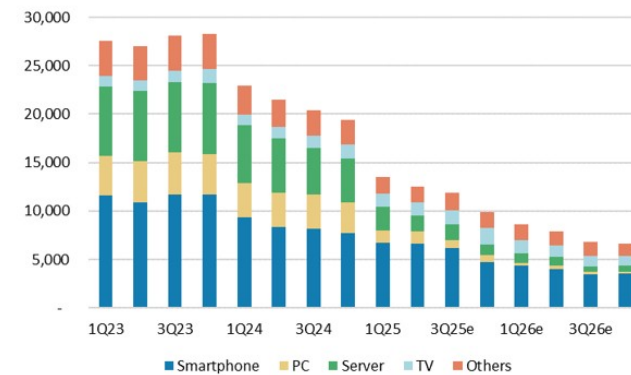
Source: ChinaFlashMarket, Morgan Stanley Research estimates.

Morgan Stanley

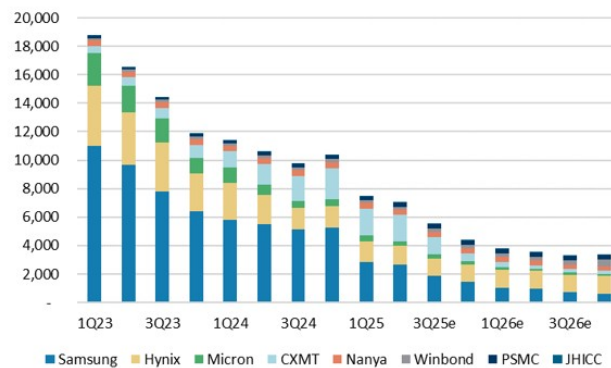
DDR4 Shortage to Persist into 2H26; Spot Price Capped



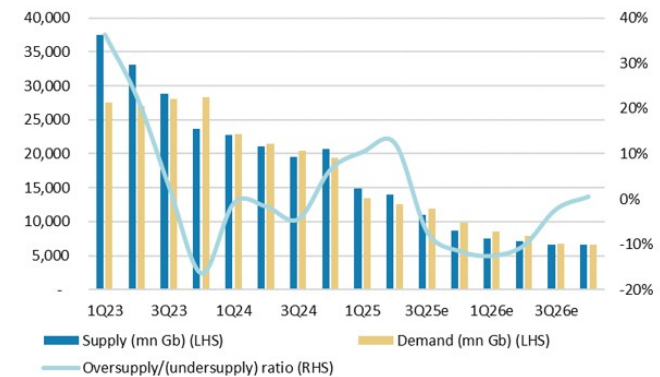
Quarterly demand breakdown, by product (mn Gb)



Quarterly supply breakdown (mn Gb)



Quarterly supply and demand summary



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

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AI ASIC and Chip Testing

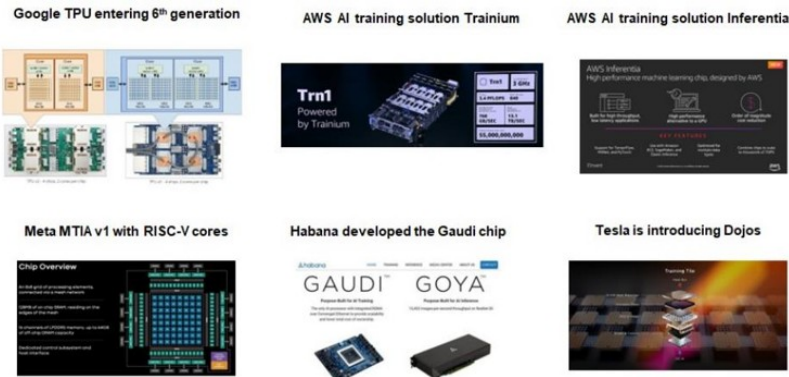
Morgan Stanley

AI Semis: Now and the Future – “Prompts”

- Generative AI Demand Driver:
 - Killer apps
 - Competition
 - Sovereign
- Growth limitations:
 1. Budget
 2. Energy -> in the US
 3. Chip Capacity -> in China
 4. Regulation
- Semiconductor solutions:
 - Moore's Law
 - CoWoS/ SoIC
 - HBM
 - CPO
 - Custom chip
 - GaN HVDC 800V
- Growth perspective:
 1. Inference vs. Training AI semis
 2. Edge vs. Cloud AI semis
 3. Custom ASIC vs. AI GPU

CSPs Still Need Custom Chips, Even with NVIDIA Providing Powerful AI GPUs

Global CSP cloud AI custom chips



The volume upside of AI chips like Google TPU could depend on ABF substrate supply



Source: AWS, Google, Meta, Intel, Tesla, Morgan Stanley Research (e) estimates.

Alchip might achieve US\$8bn revenue for Trainium4 in 2028

Alchip 2028 Trainium revenue could reach US\$8bn

		2026e	2027e	2028e
Volume (k)	Trainium3	400	600	
	Trainium4			800
Chip ASP Recognized by Alchip (US\$k)	Trainium3	4.6	4.6	
	Trainium4			10
TPU Revenue (US\$bn)	Trainium3	1.8	2.8	
	Trainium4			8
	Total Trainium Revenue	1.8	2.8	8
Total Alchip Revenue (US\$bn)		3.1	4.4	9.8
% of Trainium Revenue		60%	63%	82%

Trainium Forecasts from Our Asian Research team

k Units	2023	2024	2025e	2026e	2027e	2028e
Trainium1/Inferentia2	300	600	30			
Trainium2		300	1,200			
Trainium3			85	1,700	2,380	-
Trainium4					unknown	2,500
Total	300	900	1,315	1,700	2,380	2,500

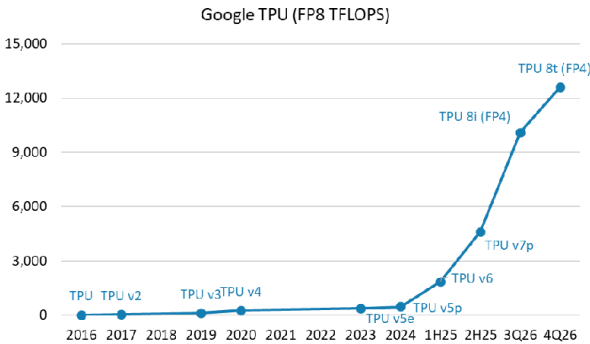
Source: AWS, Google, Meta, Intel, Tesla, Morgan Stanley Research (e) estimates.

More ASIC Projects Are Coming, According to Each CSP Plan

TPU Forecasts from MS Asia Research team

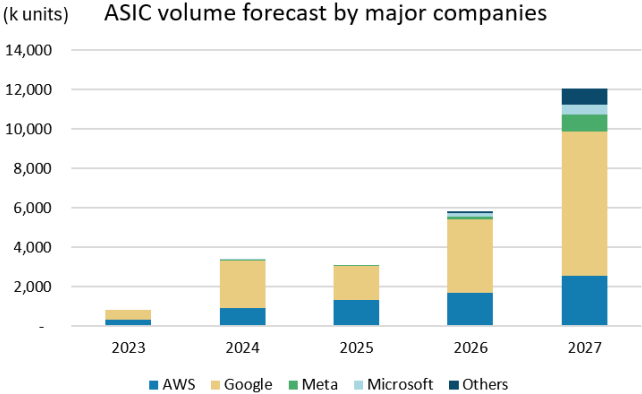
k Units	2023	2024	2025e	2026e	2027e	2028e
v5	500	2,400	250			
v6 (Trillium)			1,200			
v7 (Ironwood, by Broadcom)			300	2,300	200	
v8i (Sunfish; 3nm, by Broadcom)				900	4,000	2,500
v8t (Zebrafish; 3nm, by MediaTek)				500	3,000	1,000
v9 (Humufish; 2nm, by MediaTek)					150	3,000
v9a (Merope; 2nm, by US design service)						unknown
v10 (Icefish; 1.4nm, by MediaTek)						unknown
Total	500	2,400	1,750	3,700	7,350	>6500

Google’s TPU performance is growing rapidly



Source: Company data, Morgan Stanley Research (e) estimates

ASIC volume forecast by major companies



MediaTek TPU revenue could reach US\$70bn by 2029 even under a COT model

We believe MediaTek would still at least handle the packaging and I/O die for TPU v10

MediaTek/Intel EMIB-T advanced packaging	
Google	KGD (Known Good Die) /Compute die
Memory Suppliers	HBM Chip
MediaTek	Electrical I/O Die
Marvell or others	Memory I/O
GUC or others	3D IC/CPO

We expect MediaTek 2029 TPU revenue to reach US\$70bn

	2027e	2028e	2029e
Volume (mn)	TPU v8t (Zebrafish)	3	1
	TPU v9 (Humufish)		3
	TPU v10 (Sunfish)		1
Chip ASP Recognized by MediaTek (US\$k)	TPU v8t	4.5	4.5
	TPU v9		13
	TPU v10		18
TPU Revenue (US\$bn)	TPU v8t	13.5	4.5
	TPU v9		39
	TPU v10		18
	Total TPU Revenue	13.5	43.5
Total MediaTek Revenue (US\$bn)	35	67	NA
% of TPU Revenue	38%	65%	NA

Source: Company data, Morgan Stanley Research (e) estimates

KYEC Benefits from Strong AI GPU, TPU and CPU Growth

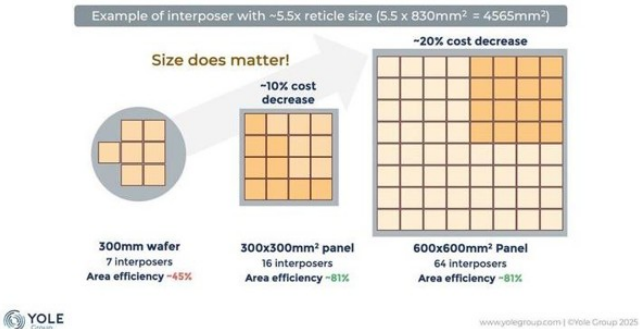
AI Revenue Forecast For KYEC

	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	2025	2026e	2027e
Units (k)											
H-series	650	200	250	500	400	400	600	600	1,600	2,000	
B-series with GDDR7	-	-	300	500	250	250	250	250	800	1,000	
B-series with HBM	1,000	1,100	1,400	1,500	1,500	1,500	1,600	300	5,000	4,900	560
R-series							500	1,500		2,000	6,500
G-series CPU									1,100	2,450	250
V-series CPU										2,500	5,750
Google TPU									1,750	3,680	4,000
Google CPU										1,500	3,000
Test time											
Per chip test time (seconds)											
H-series test time	350	350	350	350	350	350	350	350	350	350	350
B-series with GDDR7 test time			350	350	350	350	350	350	350	350	350
B-series with HBM test time	700	750	900	1,000	1,000	1,000	850	850	1,000	850	850
R-series test time							850	850		850	1,275
G-series CPU test time									300	300	300
V-series CPU test time										700	700
Google TPU blended test time									186	340	450
Google CPU test time										300	300
Total test demand (k hrs)											
H-series	63	19	24	49	39	39	58	58	156	194	-
B-series with GDDR7			29	49	24	24	24	24	78	97	-
B-series with HBM	194	229	350	417	417	417	378	71	1,190	1,282	132
R-series					-	-	118	354		472	2,302
G-series CPU									92	204	21
V-series CPU										486	1,118
Google TPU									90	348	500
Google CPU										125	250
Test hour rate (for the largest customer including burn-in)											
	170	170	170	170	170	170	180	190	170	178	200
Test hour rate (for TPU with self-bought testers)											
									200	250	280
Largest customer's AI revenue to KYEC (US\$ mn)	43.8	42.3	68.6	87.4	81.6	81.6	104.1	96.5	257.6	486.3	714.6
% of KYEC MSe Revenue (quarterly figures without CPU)	19%	15%	23%	27%	25%	23%	27%	21%	23%	32%	32%
TPU revenue to KYEC (US\$ mn)									18.1	86.9	140.0
% of KYEC MSe Revenue									1.6%	5.7%	6.2%
Google CPU revenue to KYEC (US\$ mn)										31.3	70.0
% of KYEC MSe Revenue										2.0%	3.1%

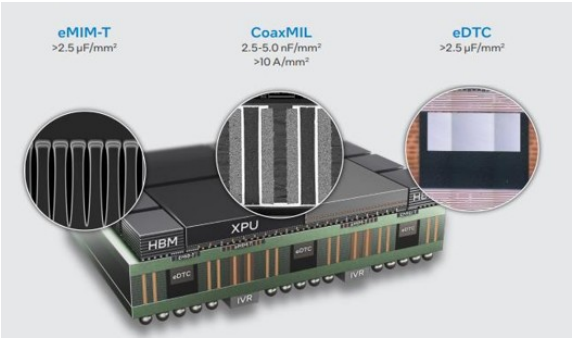
Source: Company data, Morgan Stanley Research (e) estimates

What About the Competition Between TSMC’s CoWoS and Intel’s EMIB?

TSMC CoWoS can support up to 9.5x reticles, or four chips per wafer

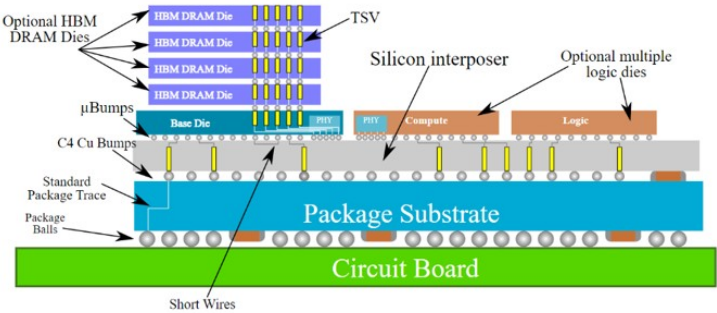


Intel EMIB eDTC: APMemory’s technology

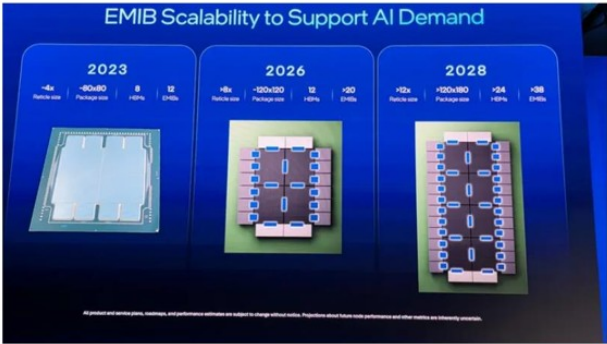


Source: Intel, Yole, Morgan Stanley Research

TSMC’s CoWoS-S (silicon interposer) packaging architecture



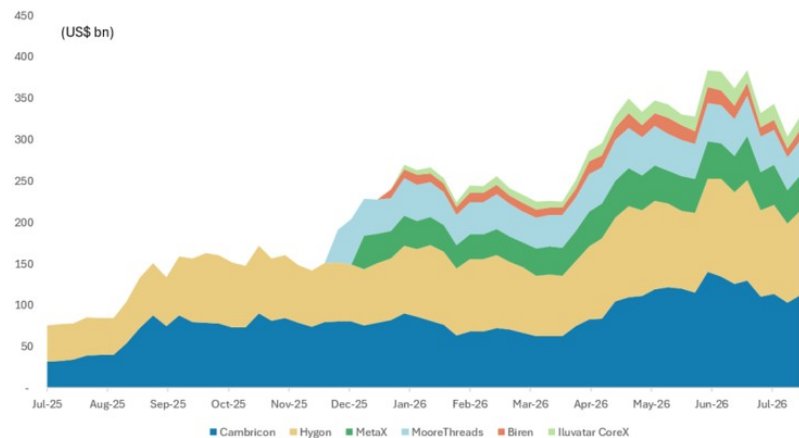
Intel’s EMIB can easily support larger chips with more reticles (>12) if its supply chain can execute well



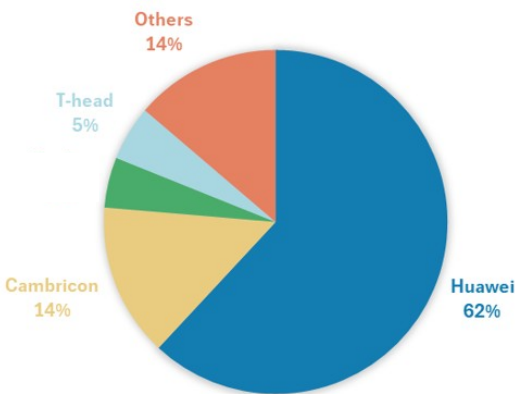
China – AI Computing

Growing Market Size and Share of Domestic AI Semis

China’s AI GPU market cap is growing, with more IPOs to come



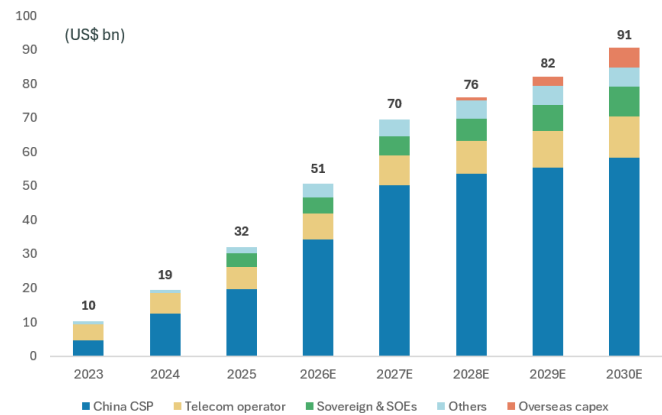
Domestic AI accelerator market share in 2026e



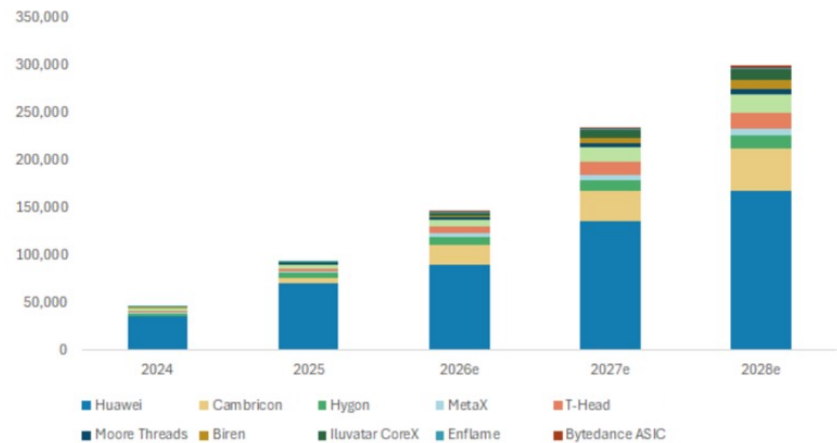
Source: Bloomberg, company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

We Expect China’s AI GPU TAM to Grow to US\$91bn by 2030e

We expect China’s AI chip TAM to grow to US\$91bn by 2030



AI accelerators' revenue trend



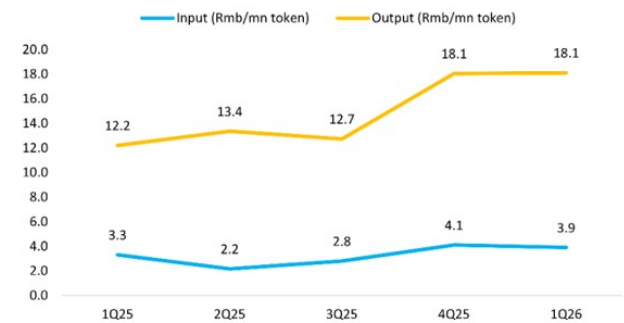
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Near-term Market Tracker for China’s AI GPU Demand

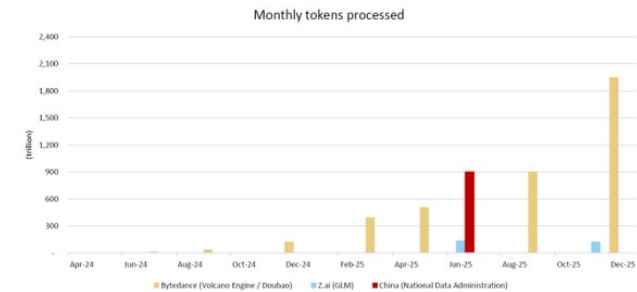
Nvidia's 5090 price keeps rising in China



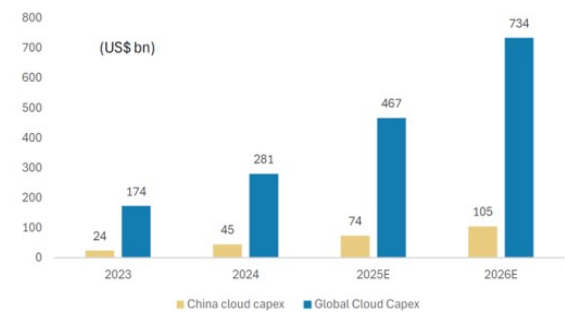
Average token price for China's mainstream AI LLMs



Surge in ByteDance (Volcano Engine/Doubao) tokens indicates high AI demand



China vs. global – cloud capex trend



Source: Company data, Taobao, Morgan Stanley Research.

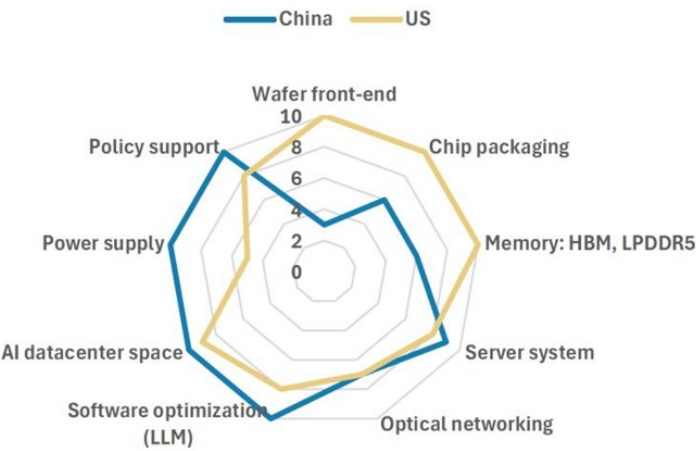
AI Chip Value Chain – China vs. the US – A Decoupling of AI Computing



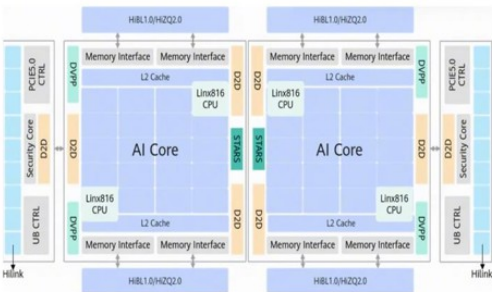
Source: Morgan Stanley Research.

China’s Infrastructure Strength Narrows Perceived Technology Gap

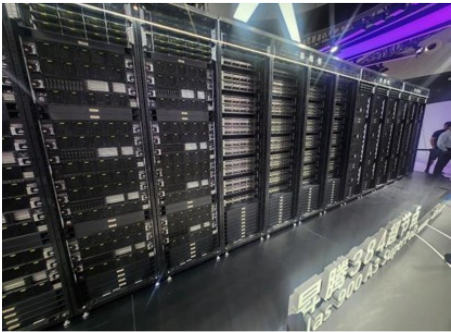
Nine factors comparison between US and China AI – Chip, System, Infrastructure



Die shot of Ascend 950



Huawei CloudMatrix 384 A3 SuperPod

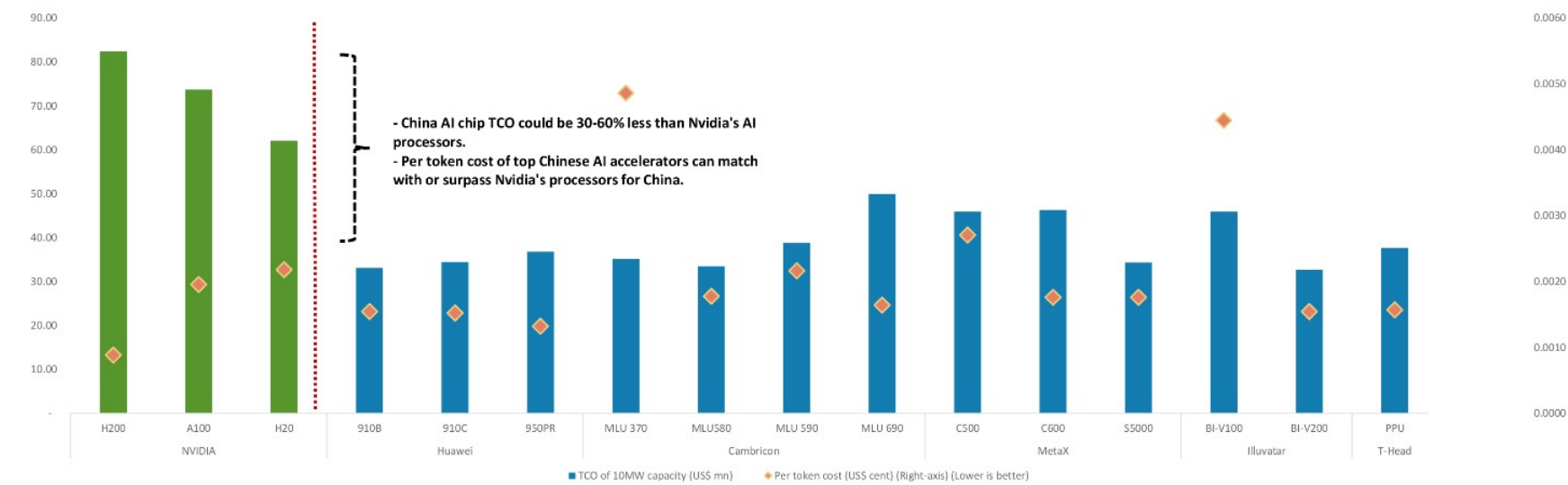


- Step 1: If one computing die is not powerful, package more dies into a single chip.
- Step 2: If one chip is not powerful enough, build larger racks and clusters.
- Step 3: If one fab is not sufficient, expand manufacturing capacity.

Source: Morgan Stanley Research.

Inference Economics: TCO and Cost per Token

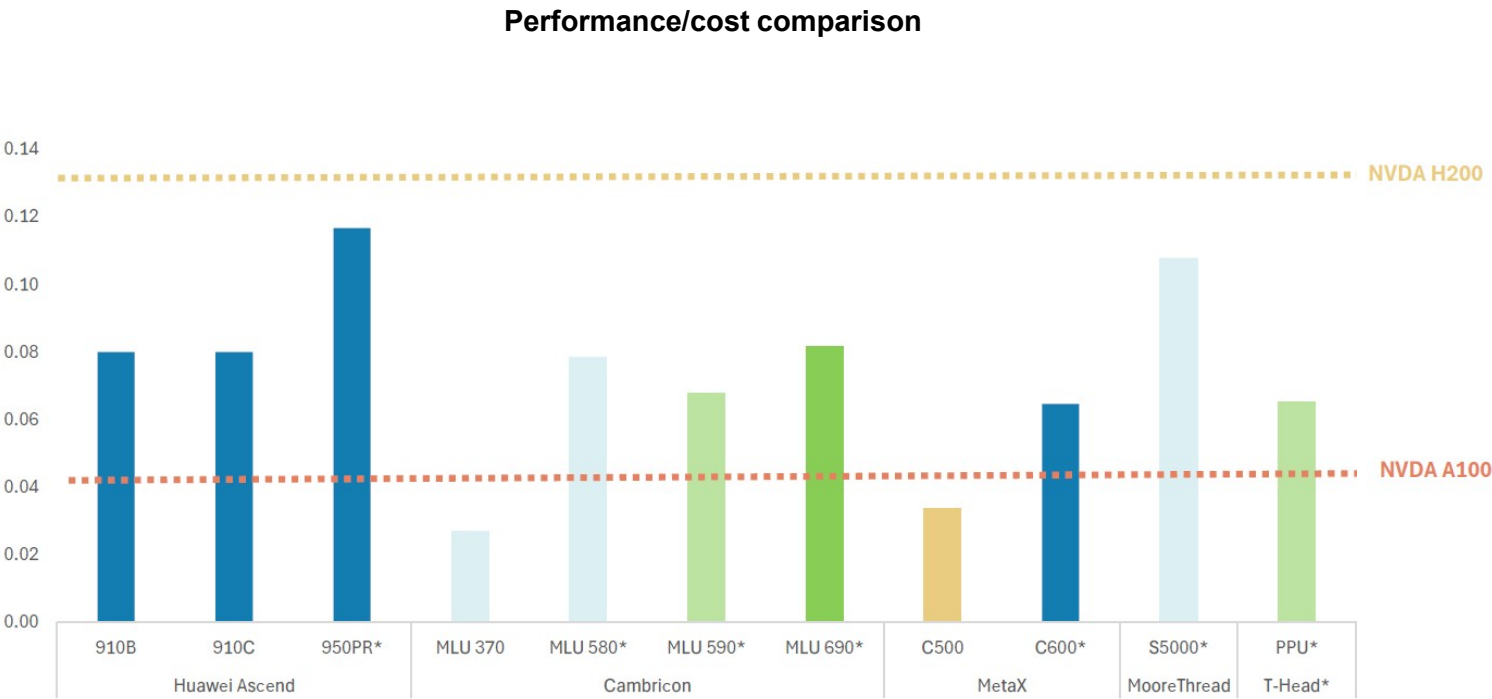
Domestic chips have lower TCO and comparable per token cost (AI LLM inference) vs. NVIDIA's processors for China



Source: Company data, Morgan Stanley Research estimates.

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Domestic Chips Deliver Stronger Performance per Dollar Thanks to Materially Lower Pricing



* = Based on Morgan Stanley estimates. Source : Company data, Morgan Stanley Research.

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Huawei Ascend Atlas 950: Scaling the SuperPod Beyond 1,000 NPUs

Huawei showcased its new Atlas 950 SuperPod at 2026 WAIC



NPU blade (tray) of Atlas 950 SuperPod



The reverse side of Atlas 950 mainly showcases the optical interconnects adopted by UBLINK



Source: Company data, Morgan Stanley Research.

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A Flourishing SuperPod Landscape at WAIC 2026

Comparison of some key SuperPod solutions at 2026 WAIC

Solution	Vendor / Partner	Scale	Architecture Highlights	Key Differentiator
Atlas 950	Huawei	1,024 Ascend 950DT NPU	16 compute cabinets + 4 UnifiedBus cabinets; UnifiedBus 2.0 unified memory semantics; hybrid copper + optical interconnect; up to 256TB pooled memory	Largest scale-up domain showcased at WAIC, emphasizing large-model training and unified memory architecture
MTT C256 SuperPod	Moore Threads	128-256 S series GPGPU	Single-level scale up with cabletree connections.	Focus on scalable domestic GPU clusters for AI training and inference
Yunsui SL64-O	Enflame + ZTE	64 L600 AI accelerators	Co-developed rack-level solution using midplane-free orthogonal architecture integrating compute, networking and cooling	Demonstrates ODM-driven ecosystem collaboration and modular SuperPod deployment
Xijing S6000	MetaX + ZTE	64 C600 AI accelerators	Similar as Yunsui SL64-o, using midplane-free orthogonal architecture	Equipped with latest MetaX AI training + inference accelerator, C600 chip

Midplane-free orthogonal designs at WAIC



Source: Company data, Morgan Stanley Research.

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Comparison Among Cambricon, MetaX, and Iluvatar

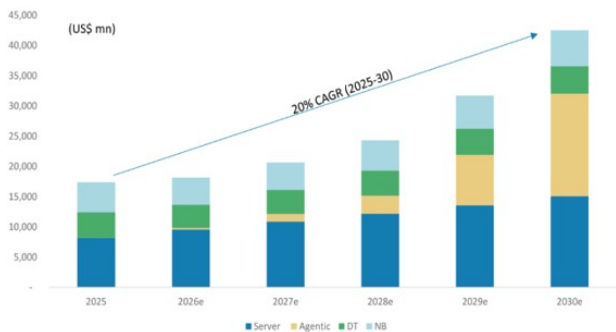
			
Ticker	688256-SH	688802-SH	9903-HK
Product	MLU 220/270/370/580/590/690 (AI Training + inference)	C Series (AI training + inference) N Series (AI Inference) G Series (Graphic rendering)	Tiangai 100/150/200/300 (AI training + inference) Zhikai 100 (AI inference)
GPGPU/ASIC	ASIC	GPGPU	GPGPU
Chip suppliers			
Process node for latest products	7nm/N+2	12nm/N+1	7nm
Secured orders from major CSPs	✓	✗	✓
Sovereign fund as major shareholder	✗	✓	✗
Per token cost performance	↑	→	→
2025 Revenue (Rmb mn)	CNY 6,497	CNY 1,644	CNY 1,034
Profitability	✓	✗	✗

Source: Company data, Morgan Stanley Research.

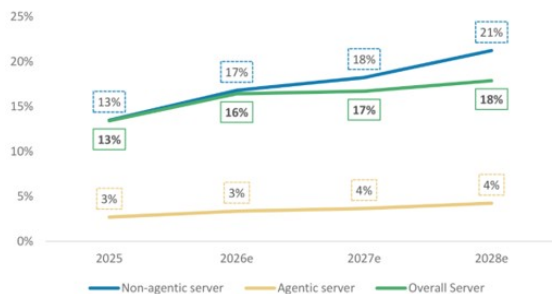
42

Hygon: Integrated CPU+GPU Compute Platform; OW

We expect China's CPU TAM to rise to US\$42bn in 2030...



...with Hygon's share in China server CPUs hitting 18% in 2028e



Source: IDC, Company data, Morgan Stanley Research estimates.

Quantum Security

Governments have established Post-Quantum Cryptography migration roadmaps

Exhibit 13: Migration timelines differ across regions, with 2030 as a key inflection point

United States	2027	Deprecate vulnerable cryptographic algorithms	2030	Complete migration of critical systems
European Union	2026	Publish migration roadmap	2030	Complete protection of sensitive information systems
United Kingdom	2028	Complete inventory and initial migration planning	2035	Full migration completed
South Korea	2028	Complete inventory and migration planning	2035	Complete migration of national cryptographic systems

Major economies are largely aligning with NIST PQC standards, while some countries continue developing domestic cryptographic frameworks.

Source: Topology Research Institute, Morgan Stanley Research

Source: Morgan Stanley Research estimates.

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Asymmetric cryptography is the dominant security framework in modern electronic systems

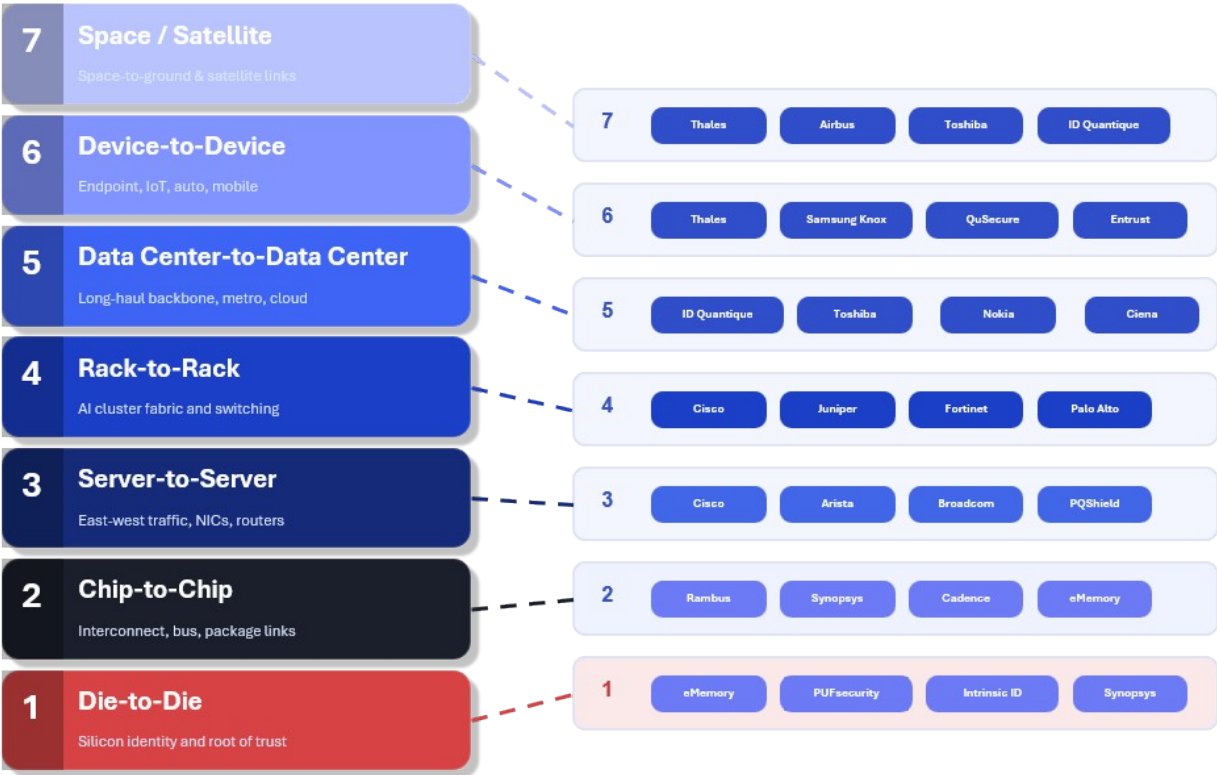
Exhibit 21: Asymmetric encryption



Source: cheapsslsecurity

Source: Cheapsslsecurity, Morgan Stanley Research estimates.

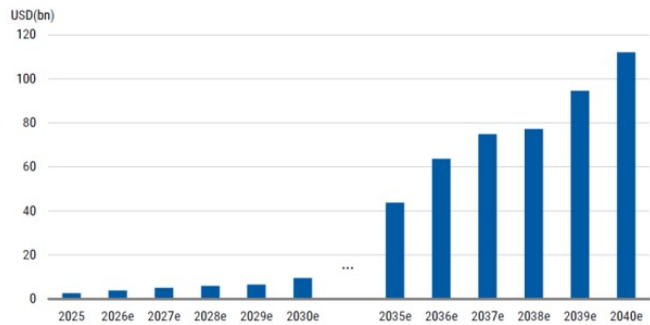
Quantum security across different connectivity layers



Source: Company data, Morgan Stanley Research estimates.

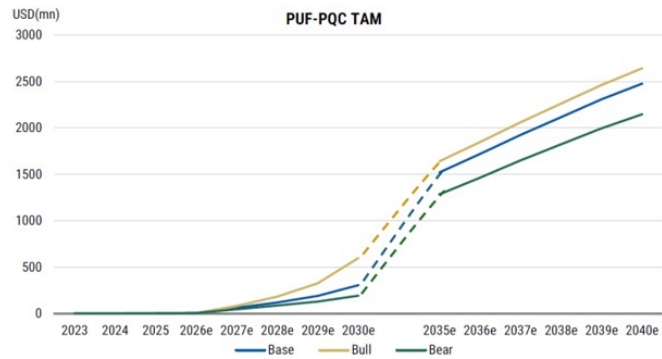
We estimate PUF TAM to be around US\$310mn by 2030

Exhibit 1: We project the quantum infrastructure TAM will reach US\$110bn by 2040



Source: Morgan Stanley Research (E) estimates.

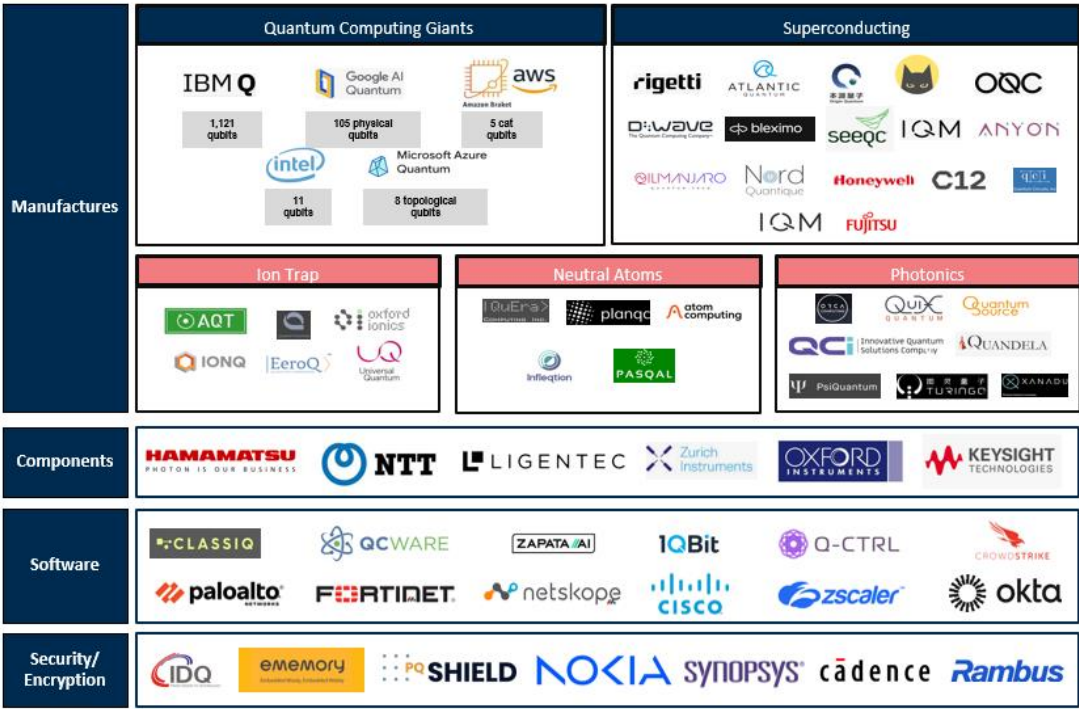
Exhibit 3: The quantum hardware security TAM – our bull, base and bear cases



Source: Company data, Morgan Stanley Research estimates.

Source: Company data, Morgan Stanley Research estimates.

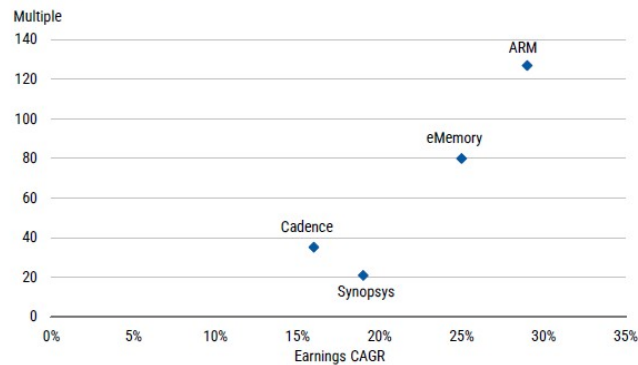
Global quantum supply chain – Hardware security is the foundation



Source: Company data, Morgan Stanley Research estimates.

Global PUF/IP comp & eMemory P/E band

Exhibit 6: Global Semi/PUF IP-related stocks – 2027 P/E vs. 2025-28 earnings growth



Source: Morgan Stanley Research estimates.

Exhibit 14: eMemory: P/E band



Source: FactSet, Morgan Stanley Research (e) estimates

Source: Company data, Morgan Stanley Research estimates.

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Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

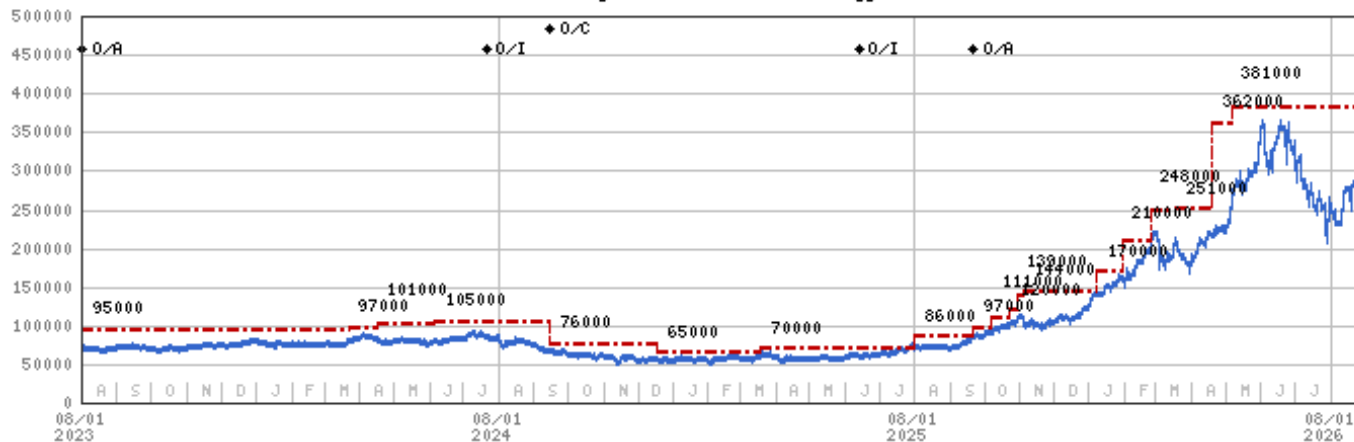
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Samsung Electronics (005930.KS) - As of 08/24/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 8/1/21 : 0/I; 8/12/21 : 0/C; 10/4/22 : 0/A; 7/21/24 : 0/I; 9/15/24 : 0/C; 6/13/25 : 0/I; 9/21/25 : 0/A

Price Target History: 6/8/21 : 98000; 8/12/21 : 89000; 9/15/21 : 95000; 12/3/21 : 97000; 3/18/22 : 95000; 4/28/22 : 85000; 6/10/22 : 80000; 7/5/22 : 75000; 7/22/22 : 70000; 9/17/22 : 68000; 3/21/23 : 70000; 5/30/23 : 90000; 7/7/23 : 95000; 3/22/24 : 97000; 4/16/24 : 101000; 6/6/24 : 105000; 9/15/24 : 76000; 12/18/24 : 65000; 3/19/25 : 70000; 8/1/25 : 86000; 9/21/25 : 97000; 10/8/25 : 111000; 10/23/25 : 120000; 10/30/25 : 139000; 11/5/25 : 144000; 1/8/26 : 170000; 1/30/26 : 210000; 2/24/26 : 248000; 3/18/26 : 251000; 4/19/26 : 362000; 5/6/26 : 381000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

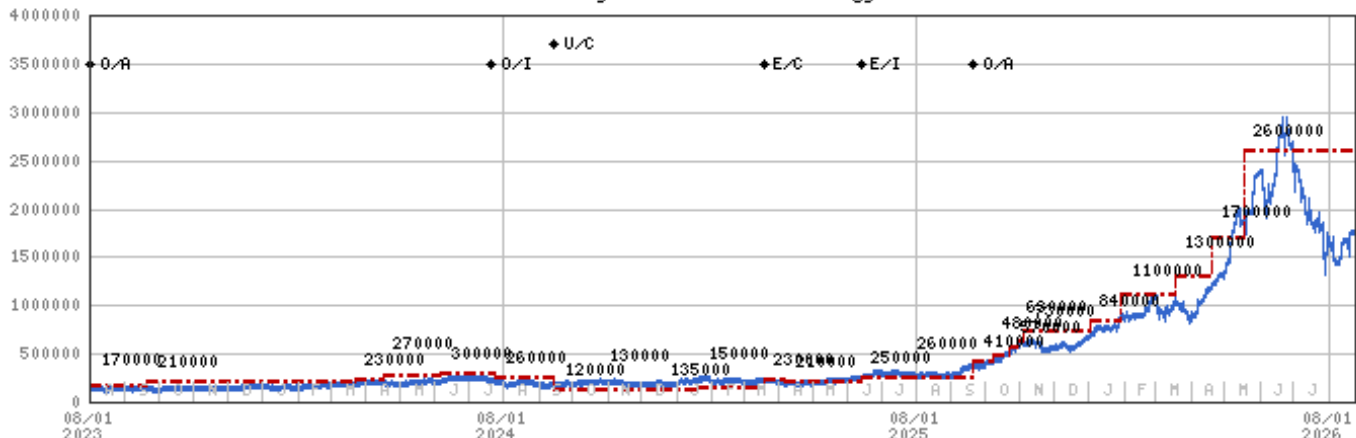
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

SK hynix (000660.KS) - As of 08/24/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 8/1/21 : O/I; 8/12/21 : U/C; 12/3/21 : E/C; 2/11/22 : O/C; 7/22/22 : E/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : U/C; 3/19/25 : E/C; 6/13/25 : E/I; 9/21/25 : O/A

Price Target History: 6/8/21 : 156000; 8/12/21 : 80000; 9/15/21 : 88000; 12/3/21 : 110000; 12/23/21 : 125000; 1/24/22 : 130000; 1/28/22 : 136000; 2/11/22 : 155000; 3/18/22 : 150000; 4/27/22 : 130000; 6/10/22 : 120000; 7/5/22 : 110000; 7/22/22 : 105000; 10/4/22 : 130000; 12/7/22 : 120000; 3/21/23 : 110000; 5/30/23 : 140000; 7/7/23 : 170000; 9/21/23 : 210000; 3/22/24 : 230000; 4/16/24 : 270000; 6/6/24 : 300000; 7/25/24 : 260000; 9/15/24 : 120000; 10/24/24 : 130000; 12/18/24 : 135000; 1/20/25 : 150000; 3/19/25 : 230000; 4/7/25 : 210000; 6/13/25 : 250000; 7/24/25 : 260000; 9/21/25 : 410000; 10/8/25 : 480000; 10/23/25 : 570000; 10/29/25 : 630000; 11/5/25 : 730000; 1/2/26 : 840000; 1/30/26 : 1100000; 3/18/26 : 1300000; 4/19/26 : 1700000; 5/18/26 : 2600000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/24/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$79.41
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb357.65
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$112.00
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$3,735.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$592.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb969.05
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT\$5,365.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$1,010.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$464.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$108.80
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$361.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$239.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb74.55
MediaTek (2454.TW)	O (11/28/2025)	NT\$3,765.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb664.00
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$501.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb707.90
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb82.50
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$2,135.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb111.13
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$450.00

SMIC (0981.HK)	O (10/21/2025)	HK\$66.75
TSMC (2330.TW)	O (02/07/2022)	NT\$2,375.00
UMC (2303.TW)	O (05/19/2026)	NT\$123.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$151.00
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$355.00
Daisy Dai, CFA		
ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$157.80
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb61.22
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb91.25
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb35.56
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb230.81
Innoscence (2577.HK)	E (10/13/2025)	HK\$43.48
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb73.80
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$25.34
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb105.05
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb92.99
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb60.99
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb25.90
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb97.80
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$845.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,400.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$15,115.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$98.90
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$139.00
eMemory Technology Inc (3529.TWO)	O (08/18/2026)	NT\$2,200.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb108.31
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb381.66
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$124.00
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$250.00
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb191.83
Novatek (3034.TW)	U (02/04/2026)	NT\$542.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$116.00
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$557.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$70.80
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$716.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb47.29
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$177.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$111.00
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$200.50
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb116.10
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb364.90
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,205.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$567.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$13.79
Hon Precision (7769.TW)	O (04/17/2026)	NT\$6,255.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$5,565.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$258.36
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$6,145.00

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

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